tasks

Knowledge Areas & Bidirectional Deep Linking — Implementation Plan

> **TASK IDS ARE PER-FEATURE AND ARE NOT GLOBALLY UNIQUE — CITE THEM QUALIFIED.** This file and [`specs/001-workshop-curriculum-platform/tasks.md`] number their tasks independently from `T001`, so the same id names two unrelated pieces of work in the two documents. **Measured 2026-09-03, the overlap is total, not an edge case: `T001` through `T120` — all 120 of feature 001's task ids — exist in both files meaning different things.** Only this file's `T121`–`T143` are unambiguous on their own. `T115` here is the answer-against-question verifier; `T115` in 001 is the `workshop/scripts/index.sh` wrapper. `T116` here is the verifier that could not decide; `T116` in 001 is the `crossref.sh` wrapper.
>
> **The requirement ids collide the same way, and by the same construction.** Measured the same day against the two `spec.md` files, which define their own ids independently from `001`: `SC-001`–`SC-021` (**21** ids) and `FR-001`–`FR-048` (**48** ids) are defined in both specs with different meanings. This feature's `SC-009` is a media-backed citation landing inside its cited span; 001's `SC-009` is citation support over generated answers. Unique to this file: `SC-015a`–`SC-015e`, `SC-017a`–`SC-017c`, `SC-022`–`SC-036`, `SC-029a`, `SC-029b`, `FR-049`–`FR-066` and `T121`–`T143`. Unique to 001: `SC-016a`.
>
> **Rule — prefix the feature number: `002:T115` vs `001:T115`, `002:SC-009` vs `001:FR-030`.** A bare `T115` in a brief, a commit message, a review or a source comment is ambiguous by construction. An unqualified citation of exactly this form already exists in the platform backend's own source at `workshop/platform/backend/`

> `pkg/answer/outcome.go`, where the pair it names is this file's; and two briefs written from these files are **reported** to have resolved a bare id to the wrong task — that report is recorded here as received and was not independently measured.
>
> **Ids are NOT renumbered to fix this.** Every existing cross-reference — in both `spec.md` files, in both `contracts/` trees, in `plan.md`, and in the implementation's own source comments — resolves against the current numbering, and renumbering would invalidate all of them at once. Qualification at the point of citation is the remedy; renumbering is not.

> **For agentic workers:** REQUIRED SUB-SKILL: use `superpowers:subagent-driven-development` (recommended) or `superpowers:executing-plans` to implement this plan task-by-task. Steps use checkbox (`- [ ]`) syntax for tracking.

Goal: turn the workshop from a media viewer into a curriculum — a knowledge layer of areas, terms, lessons and provenanced questions, joined to the transcript, the recording and the code by one bidirectional graph, searchable end to end, and produced by a pipeline the next chapter runs through unchanged.

Architecture: the knowledge layer is derived from the existing passage registry and mints its identifiers through the **same minter**, so every link inherits a survival property that is already measured rather than a new one that would have to be proven. Mentions are the join: term or area → passage → text span, and for media-backed passages a time span at one of two explicitly-labelled precisions. Areas, terms, lessons and questions join the existing two-leg search rather than arriving with a second one. The front end gains the reference module's learning vocabulary and subordinates the media views it already has.

Tech stack: Go 1.26.2 · Angular 19.2 standalone (Karma/Jasmine, Playwright) · SQLite + FTS5 · the existing identity library consumed from `submodules/passage` · fusion and reranking consumed from the existing retrieval library · Python 3.14.6 + Bash for the pipeline · a Markdown converter, an HTML-to-PDF renderer and a Mermaid renderer for export, **all user-local on this host and therefore capability-probed, never assumed**.

Global constraints

Every task's requirements implicitly include this section. Values are copied verbatim from `spec.md` and `research.md` — do not round them and do not "improve" them.

- **Shape from the reference, substance from our own chapters** (FR-004, FR-004a). The reference holds **34 finished area documents** and **785 finished questions** about a different subject, in a **private** repository. Copying them would be faster, would survive a superficial review, and would be worthless. This is the single easiest way for this feature to go wrong.
- **One minter, five new kinds** (D-KG-2). No identifier derives from text, title, slug, ordinal, heading or position. The reference's positional question code is **not** ported as a linkage.
- **The graph is keyed on the passage identifier; never on a time** (FR-033b, D-KG-5). A time-keyed graph does not break loudly — it keeps resolving and points a few seconds off.
- **Mention precision is two-valued and always declared** (FR-021, D-KG-1). Measured: 15,610 words, **15,535 (99.52%) join to a segment, 75 (0.48%) do not**; word probability p05 **0.559**, p50 **0.997**; **17.4%** below the flag threshold. Segment durations: median **6.74 s**, p95 **10.78 s**, max **20.22 s**.
- **A question without a resolving citation is withheld** (FR-036, D4). Withholding is the default path, not an error path. The reference's measured value for this property is **0 of 785**.
- **A long-set question cites more than one distinct passage** (D-KG-7). This is the structural distinction between short and long; perceived difficulty is not reproducible.
- **The match-offset field is currently a promise that is never kept.** It is declared and set unconditionally to empty; the snippet uses empty delimiters. Either populate it or remove it and state its absence (FR-029, C4.1.3).
- **Existing latency thresholds are not renegotiated by corpus growth**: suggestions ≤ 200 ms p95, results ≤ 2 s p95, ≥ 90% top-5 on a ≥20-query benchmark.
- **Every check is three-valued**: `0` fine · `1` a real problem found · `2` could not determine. A missing dependency, unreachable service or crashed helper is always `2`. This conflation has been found and fixed four times in this repository.
- **Every check owes a paired mutation proof that includes a case running the real entry point against the real tree** (FR-053, SC-027). Two proofs have already shipped here that could not fail.

- **No server-side CI anywhere in the fleet** (FR-056, SC-030). The fleet-wide gate sweeps submodules, so an attempt will be caught.
- **Redaction reaches all eight propagation targets** (FR-027). Enumerated in `data-model.md` §5.
- **No coverage threshold, no area count target** (research U2, U3). Figures are measured and published; a guessed target gets optimised toward.
- ~~**Three open clarifications bound real work.**~~ ~~**Correction, 2026-09-02: "three" is now TWO."**~~ **Correction, later the same day: "two" is now ONE, and both earlier counts are WITHDRAWN rather than restated.** The block used to read: prose authorship bounds materials (Phase 4); the on-screen-content question bounds ingestion (out of scope until answered); the answering defect bounds generation over the new kinds (Phase 10). Two of those three are now decided:
    - **Answering — DECIDED, and now BUILT.** The operator took the product decision to build the verifier, and U4 (does an entailment model load on this host?) was measured settled by T114. ~~T115 and T116 are **unbuilt**, not blocked, and their markers say `[UNBUILT]`.~~ **WITHDRAWN 2026-09-03, not restated:** both were built, wired, gated and pushed on 2026-09-02, their `[UNBUILT]` markers are discharged and both are ticked. See their own notes for the commands.
    - **On-screen content — DECIDED 2026-09-02.** The operator chose **spoken PLUS on-screen OCR**, extending this specification **in place** rather than opening a feature 003, and then chose to **specify now and build later**. It is written down as **Phase 11, tasks T123–T142**, sequenced after this specification's already-unbuilt tasks — **18 as of 2026-09-03 (second reconciliation)**, the figures of 28, then 26, then 23, then 22 this line carried being superseded by the T115/T116, T079/T083, T102 and T063/T066/T069/T117 ticks rather than wrong when written. `spec.md`'s Out of Scope entry excluding OCR is **withdrawn**; the decision and its reasoning are recorded as **D5**.
    - **Prose authorship (T031) — GENUINELY OPEN.** It is the ONE remaining clarification, and its `[BLOCKED: prose authorship]` marker is LIVE.

  The on-screen decision moved this feature's finish line, and that is recorded here rather than discovered later: spec 002 cannot close until OCR accuracy is MEASURED (SC-031 to SC-034) — not designed, not implemented, not reviewed. **Measured.** Speech-recognition accuracy in this project is a measured quantity with a recorded procedure (`workshop/pipeline/CALIBRATION.md`, `workshop/pipeline/compare_engines.py`, `workshop/scripts/verify-`

`accuracy.sh` — all three present, measured 2026-09-02). On-screen text entering the same corpus at a lower evidentiary standard than the speech beside it would make the corpus's accuracy unknown while every published figure kept looking precise.

Task format

`[ID] [markers] [Story] Description`

Markers: `[P]` parallelisable · `[TDD]` test-first · `[REVIEW]` review before proceeding · `[SUBAGENT]` delegable · `[BLOCKED]` waiting on a named clarification or unverified entry · `[UNBUILT]` the clarification or entry it once waited on is **settled** and the decision is taken — what remains is code nobody has written. Added 2026-09-02 so a settled decision can never keep wearing a `[BLOCKED]` marker, which reads as "not our move" when the move is ours.

Numbering is permanent. Ids are never reused or renumbered — other documents and agent reports cite them, so renumbering silently invalidates every citation.

Standing rules

1. **An assertion that greps for a string is not a test.** It must execute the behaviour and check the observable result. This repository has shipped an assertion that stayed green while the thing it tested was deleted.

2. **A task MUST NOT assume a capability nobody has verified.** Where a task depends on an unverified entry, it names the entry and names the task that settles it.

3. **No frozen host assumptions.** Export tooling, diagram tooling and word-timing availability are all **detected**, with environment overrides. Every export tool on this host resolves into a user-local directory.

4. **No stage writes to a source.** Sources open read-only; derived artifacts land only in derived locations.

5. **Never edit a shell script while it is executing.**

6. **Nothing from the reference module's content enters the workshop**, including as a placeholder, a fixture, or a template with the words left in.

7. **`SC-###` identifiers COLLIDE across specifications — always name the spec before citing one as acceptance.** Added 2026-09-03, after the ambiguity produced a wrong acceptance claim. The two specs number their success criteria independently and the same number means different things:

id	this spec (002)	spec 001
SC-009	a media-backed citation lands inside its cited span	100% of answer citations genuinely support the claim
SC-010	identifier survival across correction and insertion — **proven here under T053**	on ≥10 unanswerable questions, **fabricates none — NOT met**; one fabrication survives

Re-derive before citing, never from memory:

```
grep -n '^- \*\*SC-010\*\*' specs/001-workshop-curriculum-platform/spec.md \
                           specs/002-knowledge-areas-deep-linking/spec.md
```

"SC-010 is proven" is TRUE of this spec and FALSE of 001. A bare `SC-010` in a note, a commit message or an agent report is therefore not a citation — it is an ambiguity, and it has already been resolved the wrong way once.

Phase 1: Setup

Purpose: make the new packages buildable and their dependencies real.

- [x] T001 Create `workshop/platform/backend/pkg/knowledge/` and `pkg/assessment/` — **`pkg/`, not `internal/`**: Go's `internal/` is importable only from within its own module, so it forecloses reuse by language rule rather than by style, and the platform's decoupling requirements forbid it for anything reusable (FR-058)

- [x] T002 Create `workshop/pipeline/{extract,mentions,authoring}/` beside the existing pipeline stages, so the chapter-addition path **gains steps rather than being rewritten** (FR-033d, FR-033e)
- [x] T003 [P] Scaffold the four new front-end feature directories under `workshop/platform/frontend/src/app/features/{areas,practice,progress,plans}/`, mirroring the existing feature-directory conventions (FR-040)
- [x] T004 [P] Extend the capability probe in `workshop/scripts/_capabilities.sh` to **capability-probe** the export and diagram toolchain — invoke each tool with the flag that matters, never `--version`. Gate **G-KG-9** precondition (FR-014a, FR-045, FR-049)
- [x] T005 [P] Add a per-chapter word-timing detector: presence of a word sidecar is a **per-chapter** fact (contract V2), never a corpus-wide assumption (FR-021)
- [x] T006 [REVIEW] Confirm the module still adds zero CI: `git -C workshop ls-files` must show no active workflow file, and the fleet-wide gate must pass (FR-056, SC-030)

Phase 2: Foundational — identity, mentions, resolution

Purpose: the contract everything else depends on. **No user story may begin until this phase is complete**, because every later artifact points at something through the mechanism built here.

- [x] T007 [TDD] [REVIEW] Implement minting for the five new entity kinds in `pkg/knowledge/identity.go`, delegating to the **existing** minter — no second minter, no second format (contract §1 M1–M5) (FR-007)

- [x] T008 [TDD] Prove **G-KG-13** in `pkg/knowledge/identity_test.go`: run every stage twice over unchanged input and assert **zero mints**, a byte-identical taxonomy file, and no source file modified. **Paired**

mutation: make minting unconditional whenever no anchor was read this run; the gate must go red on all three assertions simultaneously (FR-009, SC-004)

- [x] T009 [TDD] Prove identifiers are neither content-derived nor positional: rename an area, fix a term's canonical form, reorder questions, shift a heading — assert **no** identifier changed. **Paired mutation**: key an area on its slug; the gate must go red. This reproduces, for the new kinds, the exact defect already measured and gated for passages (FR-007, SC-004)

- [x] T010 [TDD] Implement the mention model in `pkg/knowledge/mention.go` with **precision as a required field on every time-carrying mention** — no default, because a default makes every unjoined mention claim word accuracy (contract §3 N1) (FR-021)

- [x] T011 [TDD] Implement the word-sidecar time join in `workshop/pipeline/mentions/join.py` following contract §3 exactly: word join succeeds ⇒ `word` precision plus the word's timing confidence; word join fails ⇒ `segment` precision against the enclosing segment. **Both branches are real** — 99.52% and 0.48% measured (FR-021)

- [x] T012 [TDD] Prove **G-KG-4** and **G-KG-15**: assert every time-carrying mention declares its precision, and that the unjoined fraction is **measured and reported**, never assumed zero. **Paired mutations**: (a) omit precision and default it to `word`; (b) treat an unjoined occurrence as `word` precision at the segment start. Both must go red (FR-021, FR-053)

- [x] T013 [TDD] Implement segment-boundary spanning: an occurrence crossing a seam emits **one mention per segment it touches** (FR-022). Attaching it to one loses half its evidence, and evidence counts are what publication decisions rest on (FR-022)

- [x] T014 [REVIEW] Settle **U5** — do the 75 unjoined words cluster or scatter? Plot their time distribution against the measured silence spans. Three-valued exit. A cluster indicates a segmentation defect worth knowing about **before** mentions are built on it; scatter means the fallback branch is uninteresting. Record under `workshop/evidence/` (FR-054, FR-055)

- [x] T015 [TDD] Wire all new-kind resolution through the **one existing** resolution function with its four outcomes — found · redacted · not present · could not determine. **No second path and no fallback**: no fuzzy text match, no nearest neighbour, no prefix match, no same-hash lookup (FR-023)

- [x] T016 [TDD] Prove **G-KG-8**: delete a link target and assert a loud outcome. **Paired mutation**: make the resolver re-point by nearest text match; the gate must go red. A silently re-pointed link is the one outcome the whole identity model exists to prevent (FR-024, FR-053, SC-011)

- [x] T017 [TDD] Prove that **could not determine is never collapsed into not present**: make the registry unreadable and assert the fourth outcome. **Paired mutation**: map an unreadable registry to not-found; the gate must go red — that collapse makes a broken database look like a curriculum that never contained the passage (FR-023, FR-054)

- [x] T018 [TDD] Implement bidirectional traversal in `pkg/knowledge/graph.go`, **derived from mentions and citations** rather than from stored reverse edges. Two representations of one fact can disagree, and the disagreement has no symptom (FR-018)

- [x] T019 [TDD] Prove **G-KG-6**: inspect every stored relationship for its join key and assert none is a time. **Paired mutation**: introduce one time-keyed link; the gate must go red (FR-033b, SC-015b)

- [x] T020 [TDD] Implement cycle-safe traversal with a visited set and a depth bound, **reporting truncation** rather than stopping silently — matching how the existing cross-reference response already reports truncation (FR-026)

- [x]

T021 [P] [TDD] Extend redaction propagation to all **eight** targets in `data-model.md` §5. Gate **G-KG-7**. **Paired mutation**: skip exactly one target; the gate must go red (FR-027, SC-012)

☐

T022 [REVIEW] Review the knowledge contract implementation before anything consumes it — — **BLOCKER:** none but the work — no `phase2e-report.md` exists, and `specs/002/` has NEITHER an `analysis.md` NOR a `review.md` (spec 001 has both). This is review-record authorship, and it is one of the two things holding T122 · **OWNER: implementer/reviewer** — unblocked today every other component depends on its shape, and a shape changed after adoption is a migration (FR-005)

```
**NOT DONE (re-measured 2026-09-03, unchanged).** The brief
exists
(`workshop/docs/session-evidence/phase2e-brief.md`); **no
`phase2e-report.md` was ever
produced**, and `specs/002-knowledge-areas-deep-linking/` still
has neither an `analysis.md`
nor a `review.md` (spec 001 has both). Every consumer adopted
the knowledge contract without
the recorded review this task exists to require. **Do not
mistake
`workshop/docs/knowledge-model-contract.md` for the discharge**
— it landed 2026-09-02 in
`692a27a` at 1,372 lines and is a forward-looking *reusable
model* contract (corpus layout,
the area-document skeleton, the export pipeline, validator
invariants). It reviews nothing
that was built and names no task.
```

Checkpoint: identity, precision, resolution and traversal are proven. Stop for human approval.

Phase 3: User Story 1 — the taxonomy and its materials (P1) MVP

Goal: areas exist as first-class entities, evidenced, with materials.

Independent test: open the area list, pick five at random, confirm each has materials in the seven-section skeleton and that every substantive claim carries a citation that resolves.

☒

T023 [US1] [TDD] [REVIEW] Implement **promotion** of the five existing prose areas in `workshop/pipeline/extract/promote.py` per contract §2.1: mint identifiers, evidence each against the corpus, adopt the existing document as materials and its headings as lesson sections. **Do not rewrite the prose as a side effect of promotion** (FR-008a)

■

T024 [US1] [TDD] Implement **P2**: an area acquiring zero evidencing mentions **fails promotion loudly**. It is neither published unevidenced nor silently dropped, and both outcomes are reported. Gate **G-KG-10**. **Paired mutation**: publish one with its mentions removed (FR-008, FR-053, SC-001)

■

T025 [US1] [TDD] Implement the significance measure in `workshop/pipeline/extract/significance.py` per contract §2.2 E2/E3 — combining distinctiveness against a general-language baseline, distribution across the corpus, and corroboration in the workshop's own written material. **Raw frequency alone is disqualified**: frequency-ranking a spoken corpus returns function words. The measure **prints its inputs** so the arithmetic can be audited (FR-013, SC-002)

■

T026 [US1] [TDD] Implement area and term extraction in `workshop/pipeline/extract/derive.py`. **No area count is targeted** (E5, research U3) — the corpus evidences what it evidences. Reading the reference module from this stage is forbidden (FR-004, FR-006)

■

T027 [US1] [TDD] Implement the three reconciliation branches per contract §2.3: **R1 attaches**, **R2 adds**, **R3 contradicts ⇒ reported for a decision, neither merged nor discarded** (FR-008a, FR-033f)

■

T028 [US1] [TDD] Prove **G-KG-14**: seed a contradiction, assert it appears in the report and that neither a merge nor a discard occurred. **Paired mutation**: make R3 fall through to R1 above a similarity threshold; the gate must go red. Merging on overlap silently redraws a boundary a human drew (FR-008a, FR-033f, SC-015d)

■

T029 [US1] [TDD] Implement passage accounting per contract §2.4: `attached + classified_unattached == registry_count`, **exactly**. Publish the attached proportion as a **measured figure with no threshold** (research U2). The check is arithmetic — a total, not a ratio — so it cannot be satisfied by redefining what counts as attached (FR-010, SC-003)

- [x] T030 [US1] [TDD] Implement uncertainty marking: a low-confidence term is marked `uncertain` (FR-011), and an area evidenced **only** by uncertain passages is marked `uncertain-only` (FR-012). 267 of 1,101 passages are uncertain, so this is not a rare branch (FR-011, FR-012)

- [ ] T031 [US1] [BLOCKED: prose authorship] [REVIEW] Author area materials in the **seven-section** — BLOCKER:** **operator decision** — the prose-authorship clarification is undecided (extractive vs build-time vs run-time). 5 documents are authored with 7 `##` sections each and 489 areas serve; the earlier "host capability" excuse is WITHDRAWN. This is irreducible authorship, not engineering · **OWNER: operator**, then **human prose authorship** skeleton, **adapting the two interview-specific sections to the workshop's subject (FR-003, FR-014, SC-005).** Blocked on the first clarification, and on that alone** — extractive assembly, agent-authored at build time, and run-time generation are three different pieces of work, and nobody has decided which one this is. ~~This host has **no generative model**, so the third cannot run here today.~~ **That sentence is WITHDRAWN, not restated, and it was false when written.** Measured 2026-09-02: `ollama list` carries the generative `qwen2.5:3b-instruct-q4_K_M` (1.9 GB), and `podman inspect workshop-curriculum_platform_1 --format '{{.Config.Cmd}}'` shows `-answer-provider ollama -answer-model qwen2.5:3b-instruct-q4_K_M` in the RUNNING container's own argv — so the model is not merely installed, it is already wired into the platform's answer path. An entailment model loads on this host too, recorded under §10.12 of `workshop/docs/limits.md` as a DECIDED verdict (`entail=0.9924`). **All three options are therefore runnable here; what blocks this task is the clarification, not the host.** Re-derive with the two commands above rather than trusting this sentence — a host changes under you, and this task's text has already been wrong about it once.

```
**PARTIAL (re-measured live 2026-09-03; the figures moved and
the gap did not).** 5 area
```

```
source documents exist under `workshop/docs/training/areas/`,
each in the seven-section
skeleton; `/api/areas` now serves **495** areas at generation
**54** — re-verified later the
same day at generation **67**, still 495 served against still 5
authored `.md` documents under
`docs/training/areas/` — so **490** have no
authored materials. **The 499/494 figures this note used to
carry are superseded, not wrong
when written**: 5 areas moved into `held_back` with reason
`all_evidence_redacted` when the
chapter-01 redactions of decision 26 were applied — publication
liveness working, not
authorship progress. The prose-authorship clarification is
still open, and `pipeline/extract/verify.py` keeps `W2-claims-
over-real-materials` and
`W3-review-over-real-materials` in a separate
`MATERIALS_STATUS_CHECKS` map — outside `GATES`
and outside `--prove-failure` — naming this gap in its own
comment. **The `[BLOCKED: prose
authorship]` marker above is LIVE**, unlike the two stripped
from T060 and T094 on the same
day: the clarification it names is genuinely undecided. Do not
tick this task on the strength
of the host correction — the correction removes a false excuse,
not the blocker.
```

- [x] T032 [US1] [TDD] Implement the authored/assembled marking on every lesson section (FR-017, contract W1). Build the field regardless of how the clarification resolves — it costs one field and keeps all three options shippable without a schema change (FR-017)

- [ ] T033 [US1] [TDD] Implement claim citation enforcement (contract W2): every substantive claim — **BLOCKER: predecessor T031** — 224 of 532 claim blocks across all 5 documents are uncited or cite a non-resolving pid (48 are exempt Sources entries); W3 is reviewed for 2 of 5. Enforcement cannot land before the prose it enforces is settled · **OWNER: implementer**, after T031 carries a resolving citation **or** is visibly marked as editorial framing that is not workshop content. There is no unmarked, uncited claim (FR-015, SC-006)

```
**PARTIAL (re-measured 2026-09-03, unchanged).** Enforcement
exists and hits production
(`authorship.check_area_w2`,
`verify.py::prove_w2_requires_citation_hits_production`). It
has
```

```
been APPLIED to only **2 of the 5** authored areas — `wc -l
workshop/curriculum/publication-reviews.jsonl` is still **2**
against 5 documents under
`workshop/docs/training/areas/`. The other 3 carry the
unaudited coincidental-citation risk
that the review record itself names.
```

- [x] T034 [US1] [TDD] Implement the publication review record and its **staleness rule**: a review older than the materials it reviews **fails**, it does not warn (contract W3). Recording *“nothing to change”* is valid; skipping the review is not (FR-016, SC-007)

- [x] T035 [US1] [TDD] Implement the seven-section presence check across every area source document (SC-005). Any missing section fails (FR-014, SC-005)

- [x] T036 [US1] Implement the taxonomy source file with the same byte-stability discipline the passage registry already has — sorted, fixed key order, final newline — so a diff means a change (FR-009, SC-004)

- [x] T037 [US1] [TDD] Implement `GET /api/areas` per wire contract §3.1, including **A3.1.2**: report the count of areas held back and why. A client cannot distinguish five areas existing from eleven existing and six failing publication unless told (FR-016, FR-059)

- [ ] T038 [US1] [TDD] Implement `GET /api/areas/{area}` per §3.2, and `GET /api/terms` and — **BLOCKER:** none but the work, but the fix is in the PIPELINE not the handler — `taxonomy.py` persists only the final score and never the significance inputs; the handler is already correct and honestly reports `inputs_available:false` with `score_determined:true` over 8,507 served terms · **OWNER: implementer** — unblocked today `GET /api/terms/{term}` per §3.4 — including the significance measure **and its inputs** on the single-term response (FR-013, FR-059)

```
**PARTIAL (re-measured live 2026-09-03 at generation 54, and
AGAIN at generation 67 later the
same day — unchanged both times, over 8,537 served terms).**
All four routes serve
200, but every term in `GET /api/terms` still carries
`significance.inputs_available: false` with an `inputs_reason`
(`internal/api/terms.go:188`): `pipeline/extract/taxonomy.py`
persists only the final
```

```
significance score, and `score_determined` being `true` beside
it is what makes the gap
precise rather than a general outage. The missing work is
PERSISTING the inputs
(distinctiveness, distribution, corroboration, baseline rate
and its source); changing the
handler is not the fix. `term-significance-inputs-unavailable`
remains a registered defect.
```

- [x] T039 [US1] [TDD] Implement **A3.4.2**: a term whose last evidence is redacted is **withdrawn from the taxonomy**, not merely unlinked (FR-008, FR-027, SC-012)

- [x] T040 [US1] Add route-manifest rows and contract sections for every endpoint above (FR-059). Gate **G-KG-1**. An endpoint built but undeclared fails the server-unity verifier by construction — that is deliberate and must not be “fixed” by loosening the verifier (FR-059)

- [x] T041 [US1] [REVIEW] Human checkpoint: the reconciliation of promoted against extracted areas. A wrong merge here is a wrong curriculum, and every later artifact inherits it (FR-008a, FR-033f)

```
**TICKED 2026-09-03 ON AN OPERATOR RULING (D-36), not on a
measurement — and the distinction
is the point.** Every R3 contradiction is now disposed of by a
rule rather than adjudicated
row by row: decision D-33 ruled that title-keyword overlap is
an **evidencing heuristic, not
a boundary a human drew**, so R3 must not defend it. That rule
cleared **exactly 12** of the
13 (`R3_contradict` 13 → 1) with every other branch unchanged
to the row, and it is a *rule*,
not a list — it reads the origin string the evidence finder
itself writes, so it stays true as
the corpus moves and **stops applying by itself** the moment
any other evidencing path
supplies the overlap.
**The 13th — the large deferred cluster — is NOT resolved and
is not claimed to be.** It
remains reported, routed to a named owner, and scheduled for a
**2026-09-16** re-check. D-36
asked whether an owned, dated, tracked deferral permits this
box to tick; the operator ruled
that **it does** — a deferral that is scheduled and owned is a
resolution, not an omission.
**Read the risk with the tick:** a ticked task whose last item
```

is still open can read as
complete to anyone who does not open this note. It is recorded here rather than left implicit.
Three options were explicitly NOT chosen and all remain open, each with its cost still
unmeasured: changing the evidence finder to consult area **bodies** (blast radius unmeasured);
a **term-admission occurrence floor** (the floor value is itself a judgement); and making the
curated human register binding — that register adjudicates 20 candidate strings and **all
20 remain live term proposals**, so a recorded human decision that no code reads is still
indistinguishable from no decision.

~~**NOT DONE (measured 2026-09-02).**~~ ~~no decision record anywhere~~ ~~**883 R3
contradictions**~~ — **both halves WITHDRAWN 2026-09-03, not restated. A decision record now
exists and the population is no longer 883.** Four operator rule decisions (29–32) were taken,
applied to production code and pushed in workshop `692a27a`, with the record written at
`workshop/docs/session-evidence/phase3e-r3-decisions-29-32.md` (titled for this task) over the
census at `phase3e-contradiction-typology.md`. Measured, not read: the latest pipeline run
(`workshop/evidence/knowledge-pipeline/20260902T200902Z-acb8265b/reconcile_and_taxonomy.json`)
reports `R1=1936 attach, R2=2 add, **R3=13 contradict**, S6=136 update, RULE-2b=355
discard-as-duplicate`. RULE 1 (extraction reads the corpus, not the graph) excludes 9,144
`kg_*` rows of 11,622 — 78.68% of the file. Three gates are registered
(`T041-RULE-1-graph-kinds-excluded`, `T041-RULE-2b-discards-only-covered`,
`T041-rules-over-real-corpus`) and both `verify-r3-rules.sh` and `prove-r3-rules.sh` exit
0, the prover catching both seeded defects against the real entry point (re-run
2026-09-03).

STILL NOT DONE, and the decision record says so itself in its own §7: the human checkpoint
this task asks for has not been completed. **13 R3 rows remain — 12 type-T2 rows that are
judgements only a person may make, deliberately left un-ruled because a rule over that class
is what contract §2.3 refuses, plus 1 type-T4 mega-cluster routed to a named owner with a
re-check date of 2026-09-16** (DEFERRED, OWNED — explicitly not

counted closed). There is
still no `phase3e-report.md`. Two real costs are recorded rather than netted away: A3 fell
0.1047 → 0.0894 and "no evidencing mention" rose 3,243 → 9,017, because the 9,144 `kg_*` rows
now evidence nothing. Tick this only when the 12 judgements are recorded and the T4 deferral
is resolved — the rules mechanised the 98.7% that was one artifact; they did not make the
remaining judgements.

**The residual judgements are now written as answerable decision packets (2026-09-03), and
they REGROUP: `workshop/docs/session-evidence/t041-decision-packets.md`.** A count is not a
question, so the 13 rows were tested for a shared root and they have one, measured **12 of
12 with no residual**: `promote.default_evidence_finder` evidences a promoted area by
whole-word matching of the area's **TITLE KEYWORDS** (`promote.py:224-263`), so on every T2
row the overlap passage carries an area title keyword and the outside passage carries none —
and **11 of the 12 member terms occur zero times in the prose of any area they are reported
as contradicting**. The 12 rows are therefore one question asked 12 times, not 12 questions.
13 individual adjudications become 4 decisions (D-33 evidencing heuristic, D-34 term
admission floor, D-35 the human-curated artifact register — measured **20 of 20** rejected
surface forms still live as terms — and D-36 whether the owned T4 deferral lets this tick).
Each packet carries options with measured consequences, evidence both ways, a recommendation
and what it unblocks. **The packets are QUESTIONS: nothing was decided, no rule applied, and
this task stays `[ ]`.** Re-derived the same day: production `R3_contradict` **13** (12 T2 +
1 T4) and pre-RULE-1 `R3` **1153** with branch `B1` exactly **883**, so the `883` above is
confirmed as a subset and never the total; `established_carried_in` is **495** (not 494) and
production `S6_update` is **138** (not 136). Do not quote packet content into this public
repository — it is cited here by path only.

**D-33 IS ANSWERED (operator decision 33, 2026-09-03) AND THE RULING IS IMPLEMENTED. The
sentence above — "tick this only when the 12 judgements are recorded and the T4 deferral is
resolved" — is SUPERSEDED IN ITS FIRST HALF ONLY, and the

second half still stands.** The ruling: **a title-keyword match is an evidencing heuristic, not a boundary a human drew, so R3 should not be defending it.** The 12 rows are therefore not 12 recorded judgements; they are one rule, applied.

RULE 3, in `pipeline/extract/reconcile.py` (`_is_title_keyword_artifact`, `_heuristic_match_keyword`, `HEURISTIC_EVIDENCE_ORIGIN_PREFIX`). An R3 contradiction is a **title-keyword artifact**, and therefore not a contradiction, when — for **every** promoted area it overlaps — every overlap passage is evidence for that area **only** through a `promotion-keyword-match:<kw>` mention **and** no matched `<kw>` is itself one of the proposal's own member terms. It names no row, area, term or passage; it reads the origin string `promote.default_evidence_finder` itself writes rather than re-running the match, so it stays true as the corpus moves and **stops applying by itself** the moment any other evidencing path supplies the overlap. **A hardcoded set of row ids was explicitly not built.**

What it disposed of, measured on an unchanged corpus (`md5sum` identical before and after; the read-only `analyze_r3_contradictions.py --exclude-graph-kinds`): `R3_contradict` **13 → 1**, `RULE_3_keyword_artifact` **12**, and `R1_attach` **1936**, `R2_add` **0**, `S6_update` **138**, `RULE_2b_discard` **355** all **unchanged to the row**. The 12 cleared are **exactly the 12 T2 rows** (`T2…|RULE-3-keyword-artifact: 12`, `T4…|R3-reported: 1`). **The rule was not tuned to hit 12:** the T4 mega-cluster is retained because **15 of the 17 promoted title keywords are among its 3603 member terms**, so it is a proposal genuinely about what the areas are titled after. The disqualified blanket form — origin without relatedness — clears **13**, taking the routed mega-cluster with it, which is why it is disqualified rather than simpler.

§1.1 paired mutation, and it proves the rule is not a switch-off. `verify.PROOFS["RULE-3-title-keyword-overlap-is-not-a-boundary"]` builds its promoted evidence with the **real** `promote.default_evidence_finder`, shows a contradiction that is

genuinely NOT a keyword artifact **still fires** in both ways it can be genuine (the matched
keyword IS a member term; the overlap is evidenced by a non-keyword mention), then seeds each
disqualified predicate into the **real** `reconcile` module and requires
`check_rule_3_artifact_is_keyword_only` to catch it — RED (a) blanket form, RED (b)
suppress-everything, both caught. `platform/gates/prove-r3-rules.sh` MUTATION 3 repeats the
seed against a throwaway copy of the package. Measured 2026-09-03: `verify-r3-rules.sh` and
`prove-r3-rules.sh` both exit **0**; the extraction suite is **278 tests OK** (was 264);
`verify-check-registry-002.sh` exits **0** at 47 checks with the new
`T041-RULE-3-title-keyword-artifact` row; `verify-check-registry-001.sh` **20 PASS / 0 FAIL /
1 DEBT**, unaffected. Nothing is merged and nothing is silently dropped: a cleared proposal
becomes a `keyword-artifact` taxonomy row carrying the matched keywords and both passage
sets, is threaded into §2.4 A1 accounting with a reason naming the rule, and the Go loader
learned the row type in the same change.

**THREE OPTIONS WERE EXPLICITLY NOT CHOSEN AND EACH REMAINS OPEN — none is closed by this
decision:** (1) changing the evidence finder to consult area **bodies** (packet D-33-A/B; its
blast radius across R1/R2/S6/RULE-2b is still **UNMEASURED**), (2) imposing a **term-admission
occurrence floor** (D-34; the central unknown — how many of the 4358–5101 terms a floor would
drop are real — is still **unmeasured**), (3) making the **curated human artifact register
binding** on the pipeline (D-35; 20 of 20 rejected surface forms are still live terms).

**T041 STAYS `[ ]`, and the reason is precise: the T4 row still blocks and D-36 is
unanswered.** RULE 3 disposed of the 12 T2 rows; the single remaining R3 row is the T4
over-merged mega-cluster, still reported, still routed to a named owner with a re-check date
of **2026-09-16**, still tracked as the `area-term-over-generation` defect. Packet D-36 asks
whether that dated deferral permits this checkpoint to close; **the operator has not answered
it**, and ticking on the strength of D-33 alone would answer D-36 by implication. **Do not
tick this task until D-36 is answered on the record.**

- [ ] T042 [US1] [REVIEW] Record the content boundary check **in both directions** before anything is — **BLOCKER:** none but the work — the SC-029a INBOUND half is unbuilt and the `sc029a-not-built` row is live; §10.14 records that the naive probe was rejected, so this needs a long-shingle design rather than a quick check · **OWNER: implementer** — unblocked today, design work required published (SC-029, SC-029a) (FR-004a, FR-057, SC-029, SC-029a)

  ```
  **PARTIAL (re-measured 2026-09-03, unchanged).** The outbound half exists —
  `pkg/assessment/boundary.go` with G-KG-16, plus the umbrella's
  `scripts/verify-content-boundary.sh`. The **inbound** half (SC-029a) is still NOT built:
  `platform/gates/defects-registry.tsv` still carries the `sc029a-not-built` row, and
  `docs/limits.md` §10.14 still records the naive probe attempted and rejected (1,410 of 1,438
  files matched, ~37,000 noise occurrences) with the shape a real check would need — long
  shingles plus a generic-content filter, not short-term substring matching. Same gap as T122.
  ```

Checkpoint: areas exist, are evidenced, have materials and a recorded review. Stop for approval.

Phase 4: User Story 2 — deep linking (P2)

Goal: any point reaches every related point, both ways.

Independent test: enumerate every relationship, traverse it in reverse, confirm the origin comes back. No interface required.

- [x] T043 [US2] [TDD] Implement `GET /api/areas/{area}/evidence` per §3.3, with **precision required** on every time-carrying entry and the word's timing confidence carried where precision is `word` (FR-020, FR-021, FR-059)

- [x] T044 [US2] [TDD] Implement **A3.3.3**: a redacted passage contributes no mention and the omitted count is reported, matching the existing cross-reference behaviour (FR-027, SC-012)

- [x]

T045 [US2] [TDD] Implement `GET /api/passages/{pid}/knowledge` per §3.7 — the reverse direction, and the endpoint that makes the recording navigable (FR-018, FR-019, FR-059)

- [x] T046 [US2] [TDD] Implement **A3.7.1**: every entry is reachable in **on e** step. A response that returns identifiers a client must resolve separately does not satisfy FR-019 (FR-019)

- [x] T047 [US2] [TDD] Implement **A3.7.3**: an unattached passage says it is unattached **and why**, rather than returning an empty list. “No areas” and “not yet classified” are different facts and an empty list reads as the first (FR-010)

- [x] T048 [US2] [TDD] Implement `GET /api/graph/traverse` per §3.8 across all four content kinds (FR-033a, FR-059)

- [x] T049 [US2] [TDD] Implement **A3.8.3**: a hop whose target cannot be resolved **reports its outcome and continues**; it is never dropped, because a dropped hop is indistinguishable from a hop that never existed (FR-023, FR-024)

- [ ] T050 [US2] [TDD] Implement the six-row connectivity matrix (FR-033a) and its exercise harness. — **BLOCKER:** none but the work — the gate exits 0, but `rows_implemented` (`graph_traverse.go:200`) still EXCLUDES row 4’s cross-reference-graph half, so a green gate is covering five of six rows · **OWNER: implementer** — unblocked today, one missing half-row **A row with zero exercised origins fails** — an unexercised traversal is unmeasured, not passing (SC-015a) (FR-033a, SC-015a)

```
**PARTIAL (re-measured 2026-09-03, unchanged).** `platform/
gates/verify-connectivity-matrix.sh`
still exits **0**, but the handler's own live
`derivation.rows_implemented`
(`internal/api/graph_traverse.go:200`) still reports rows
**1,2,3,5,6** plus only row 4's
shared-area-membership half. Row 4's **cross-reference-graph**
half is not implemented
(`docs/limits.md` §10.7, defects row `connectivity-row4-half-
implemented`). The gate gained a
real paired proof on 2026-09-02 (`prove-connectivity-
matrix.sh`, 7 mutations including an
```

rc=2 control) — that closed a §1.1 debt, not this task's
missing half-row.

- [x] T051 [US2] [TDD] Prove **SC-008** over the **whole** relationship set, not a sample: a one-way link is indistinguishable from a two-way one when read forward, so sampling cannot find it (FR-018, SC-008)

- [ ] T052 [US2] [TDD] Prove **SC-009**: every media-backed citation lands inside its cited span, — **BLOCKER: none but the work, and it is now a live REGRESSION rather than a proof debt** — the gate exits **rc 2**, not rc 0: `UNDETERMINED: GET /api/passages/01M1ET0MFM0EYA5TACY1R1JWEQ -> HTTP 410`. A cited passage has since been REDACTED and the gate has no branch for a 410, so ZERO assertions ran. It also still owes its paired mutation · **OWNER: implementer** — unblocked today; highest urgency of the 002 set **and the precision split is published** alongside the pass rate. A 100% pass at segment precision and at word precision are different products, and a test that only asserts “inside the span” cannot tell them apart (FR-021, SC-009)

**PARTIAL — but the blocker this note named is DISCHARGED. "No gate anywhere asserts that every
media-backed citation lands inside its cited span over the whole set … that gate is the work"
is WITHDRAWN, not restated.** The gate was built later on 2026-09-03, while this reconciliation
was in progress, and it passes. Run here rather than quoted:

```bash
bash workshop/platform/gates/verify-sc009-citation-span.sh    # rc=0, live against :8087
```

It measures **two full populations, neither sampled**: **96 of 96** served media-backed
citations across all 495 advertised areas land inside their cited span (95 distinct passages),
and **3 of 3** authored `cite: PID` markers in the five area documents do too. The precision
split is published with the result rather than smoothed — **word 85, segment 11** over the
cited subset — and the gate prints the cited subset's own segment distribution (n=96, median
7.29 s, p95 9.94 s) **next to** this spec's corpus-wide figures while stating that the two are
different populations and neither substitutes for the other. The gate also opens by naming the

SC-009 collision itself — spec 001's SC-009 is the benchmark-composition floor in
`bench-answers.sh` — which is standing rule 7 being enforced by an instrument rather than by a
reader's memory. Its own declared boundary: the recording-seek agreement (95 of 95) is **not
independent today**, because the seek is derived from the span, and it says so rather than
counting as corroboration.

Why this still does not tick. This is a `[TDD]` task, and this file's global constraint is
that **every check owes a paired mutation proof**. This gate has none: `bash
platform/gates/verify-sc009-citation-span.sh --prove-failure` returns rc=2 `unknown argument`,
and its `check-registry-002.tsv` row `T052-sc009-citation-span` is 4 fields with no paired-proof
column. A gate nobody has driven red is not yet known to be able to go red. **That single item
is the remaining work** — the measurement itself is done.

**On the `precision` FIELD, and a correction to an earlier reading of it made in this file on
2026-09-03 and withdrawn the same day.** Workshop commit `ff90e09` wired the word-sidecar
join, so the field is no longer pinned to `segment`: measured live on the `/evidence` route,
`precision: "word"` is now the common value (first 12 areas in list order → word 12, segment
1, no_time_span 4). **An earlier revision of this note read that as "word precision is live"
and drew the conclusion that `docs/limits.md` §10.1 was stale. That conclusion was WRONG and
is withdrawn, not restated — the label moved and the delivered precision did not.**

What the join actually returns, read from `pkg/knowledge/wordjoin.go` rather than inferred
from the label: `JoinSegmentWords` classifies *"its whole word range"* (its own doc comment)
and returns `StartS: sub[0].StartS, EndS: sub[len(sub)-1].EndS` — the first word's start to
the **last** word's end **of the entire segment**. It is a word-timing-derived tightening of
the segment's own bounds, not a pointer at one word. The `precision` flag flips to `word`
when every word of the segment carries a timing inside the segment window — a statement about
sidecar completeness, not about span width. Measured over 40 areas:

```
precision="word"     n=40  min=2.040s  median=7.650s
max=13.040s
precision="segment"  n= 3  min=6.300s  median=7.620s  max=
9.620s
```

```
**The two are the same width.** Both match this file's own
recorded segment distribution
(median 6.74 s, p95 10.78 s). A word's span is a fraction of a
second.

**So §10.1's substance stands and must NOT be "corrected" away:
every deep link in this
deployment still resolves at segment width, and "jump to the
exact moment a term was said"
is still "jump to the segment it was said in."** What is now
inaccurate in §10.1 is narrower
than a staleness — its *mechanism* sentence (`entry.Precision =
PrecisionSegment` set
unconditionally; `precision_split` always `{"word": 0,
"segment": N}`) no longer matches the
code. **That makes the response shape worse, not better, and it
is a NEW defect rather than
a closed one:** an entry labelled `word` carrying a 7.65 s
median span overstates its own
precision to any client that trusts the field, which is the
exact misreading §10.1's
consequence paragraph was written to prevent. Recorded here;
fixing §10.1's wording, the
`precision-segment-only` row, and the label itself is
`workshop/`-owned and out of this
file's scope.

**None of this ticks the task** — but the reason has CHANGED,
and the old one ("the SC-009 gate
does not exist") is **WITHDRAWN, not restated**. The gate
exists and is green over both full
populations; what it still owes is a paired mutation. See the
head of this note.
```

- [x] T053 [US2] [TDD] Prove **SC-010**: replay the identifier-survival experiment extended to all five new kinds; compare the full link set before and after **by hash** (FR-025, SC-010)

- [x] T054 [US2] [P] Add route-manifest rows and contract sections for the endpoints in this phase (FR-059)

- [x]

T055 [US2] [P] [SUBAGENT] Build the transcript-to-knowledge affordance in the existing transcript view — the entry point for “what is being taught at this moment” (FR-019, FR-041)

**DONE. This task was already ticked while carrying a "NOT DONE" note — the note was the stale
half, and it is WITHDRAWN, not restated.** Every specific it asserted is now false, checked one
by one on 2026-09-03 (source): `platform/frontend/src/app/core/api.ts` calls
`` `/api/passages/${encodeURIComponent(pid)}/knowledge` `` (it does not stop at `crossrefs`);
`features/transcript/transcript.component.ts` carries **6** `knowledge` references, not zero,
and renders `<app-passage-knowledge [pid]="pg.pid" />`; and the component the old note implied
did not exist is `features/transcript/passage-knowledge.component.ts`, whose header names T055.
Line numbers are deliberately omitted — both files moved under this reconciliation on the day —
so re-derive with `grep -n knowledge` rather than trusting a citation.

Proven by execution, not by reading the source. The Playwright config deliberately declares
no `webServer`, so the suite exercises whatever the container is serving rather than a dev
server it started itself:

```bash
cd workshop/platform/frontend && npx playwright test e2e/passage-knowledge.spec.ts
# 16 passed (24.9s) (served) — 8 tests × desktop-chromium + mobile-chromium
```

Those 8 include a **paired mutation** — leaving the disclosure closed must result in no call to
the endpoint at all — a check that what is rendered equals what the endpoint returned, and one
case per resolution outcome (`unattached`, `redacted`, absent from the registry, unreadable),
so the four A3.7.2 outcomes are each shown to render **distinctly** rather than collapsing into
a single empty list.

T056 [US2] [REVIEW] Review the link model against the contract before the surface consumes it (FR-005)

Checkpoint: the graph is complete and bidirectional, and both precisions are honest.

Phase 5: User Story 3 — search over everything (P3)

Goal: all four content kinds searchable, every hit says where.

- [x] T057 [US3] [TDD] Index the four new kinds — `area`, `term`, `lesson_section`, `question` — indexing **their own text**, not only the passages they cite. A question findable only through its citations is not findable by anyone who does not already know the answer (D-KG-10) (FR-028, SC-013)

  ~~~
  **3 of 4 kinds indexed; `lesson_section` is a recorded gated REFUSAL, not a gap.** The note
  this task carried — *"`corpus.indexed_kinds` still returns exactly `[transcript_segment,
  doc_section, code, diagram, area, term]` … `lesson_section` and `question` are **not
  indexed** — **2 of the 4** kinds this task names"* — is **WITHDRAWN, not restated**: `question`
  was indexed on 2026-09-03 and the live block now advertises seven kinds. Re-measured live:

  ```bash
  curl -s 'http://127.0.0.1:8087/api/search?q=quonsari&limit=2' | jq .corpus.indexed_kinds
  # ["transcript_segment","doc_section","code","diagram","area","te
  # (served)
  ```

  **`question` is indexed on its own text, which is what this task asks for.** Counted by
  unioning the pids returned for `kinds=question` across twenty probe queries (served): **44** distinct
  entries reachable through search, exactly equal to the **44** questions the API actually serves
  (9 short + 35 long across 5 areas of 495; 0 withheld). `pkg/search/question_catalog.go`'s
  `BuildQuestionEntries` runs the SAME `assessment.ServeQuestions` decision the §3.5 route runs,
  so a question whose citation stops resolving is withheld from **search** as well as from the
  practice set — G-KG-2 reaches the index rather than stopping at
  ~~~

the endpoint. Note the
mechanism, because it is not what the registry would suggest: the catalog is built from the
question bank directly, and `curriculum/passages.jsonl` mints **zero** `kg_question` records.

`lesson_section` is the fourth kind, and it is deliberately NOT indexed. Measured over the
11,622-record registry:

```bash
python3 -c '...collections.Counter(kind)...' < workshop/curriculum/passages.jsonl
# 11622 total (source): kg_term 8553 · doc_section 1172 · transcript_segment 1055 · kg_area 516 ·
#              code 251 · kg_todo 27 · kg_next_point 24 · kg_open_question 14 ·
#              kg_meeting_note 10       →  kg_lesson_section: 0
```

Zero of 11,622, and the reason is structural rather than unfinished work:
`pkg/knowledge/materials.go`'s `LessonSectionMeta` carries `Heading`, `Ordinal`, `Authorship`
and `CitationCount` and **no identifier**, so there is no key to index a row on or to join a
question to. The refusal is pinned by a test — `pkg/search/question_search_test.go:205
TestT057_LessonSectionIsNotAdvertisedBecauseNothingIndexesIt` — which fails if `lesson_section` is ever added to `corpus.indexed_kinds` without something
behind it, and which also asserts the three backed kinds ARE advertised so it cannot pass on a
build that quietly stopped advertising everything.

Why this stays unticked: the task names four kinds and three are indexed. Closing it needs
either a minted `lesson_section` identity (a data-model change) or an amendment narrowing the
task to the three kinds that can carry one. **That is a spec decision and it is not made here.**

**[TICKED 2026-09-04. The blocker note is WITHDRAWN BY NAME: the clause *"That is a spec
decision and it is not made here"* is STALE — the decision HAS been made and is written into
the contract.** `contracts/http-api-delta.md` C4.1.1 now reads *"`area`, `term` and `question`
join the existing kinds"*, with an explicit blockquote recording that **`lesson_section` was
REQUIRED here and is WITHDRAWN, 2026-09-03, by operator

decision** — 0 of 11,622 records
carried the kind and its metadata type has no identifier field, making it structurally
unkeyable. `grep -c lesson_section spec.md` = **0**, so the spec never required it either.
Measured live 2026-09-04 at generation 68, `GET /api/search?q=area` returns
`corpus.indexed_kinds = [transcript_segment, doc_section, code, diagram, area, term,
question]` — the three contracted kinds are indexed and `lesson_section` is correctly ABSENT
rather than advertised-empty, which is the zero-entry-kind defect C4.1.1 forbids. The refusal
is test-pinned, not incidental: `TestT057_LessonSectionIsNotAdvertisedBecauseNothingIndexesIt`.
`go test -count=1 -run 'TestT057|TestT058|TestGateKG11' ./pkg/search/...` -> **ok, rc 0** (in_process).
Honest boundary (§11.4.6): the contract amendment is an UNCOMMITTED working-tree change
(` M contracts/http-api-delta.md`). The tick is sound; the change is not yet committed]**

■

T058 [US3] [TDD] Advertise the new kinds in the corpus block, and prove **G-KG-11**: an advertised kind must be **retrievable**, proven by a planted known target — **not** by a row count, which cannot distinguish a populated index from a populated table nobody queries. The corpus already advertises one kind with zero entries (FR-028, SC-013)

**3 of 4 kinds advertised and proven; `lesson_section` is a recorded gated REFUSAL, not a
gap.** The note this task carried — *"its scope is `area` + `term` only. `lesson_section` and
`question` are still neither advertised … nor proven"* — is **WITHDRAWN, not restated**.
G-KG-11 now covers a third kind, proven the way the task demands — by a **planted known target
actually retrieved**, never by a row count:

- `area` + `term` — `pkg/search/knowledge_search_test.go:93`
  `TestGateKG11_AdvertisedAreaTermKindsAreRetrievable`, with its negative twin at `:176`.
- `question` — `pkg/search/question_search_test.go:75`, whose own comment is *"advertised
  BECAUSE it is retrievable"*, and which plants a question and retrieves it.

```bash
cd workshop/platform/backend && go test -count=1 -run 'TestT057|TestT058' ./pkg/search/...  # ok (in_process)
```

````
```

Live, every advertised kind was probed by unioning search
results across fourteen queries (served):
`area` **5** entries · `term` **382** · `question` **44** ·
**`diagram` 0**. The zero-entry
kind this task names in its own last sentence is therefore
**still `diagram`, and still
exactly one** — this phase added a kind that is backed and did
not add a second empty one,
which was the constraint C4.1.1 imposed. `lesson_section` is
absent from
`corpus.indexed_kinds` entirely rather than advertised empty,
which is the correct handling
and is pinned by
`TestT057_LessonSectionIsNotAdvertisedBecauseNothingIndexesIt`.

**Why this stays unticked:** it is bound to T057 — the fourth
kind is neither indexed nor
advertised, and the same spec decision settles both.

**[TICKED 2026-09-04, on the same operator amendment as T057 —
see that task's evidence.**
The corpus block advertises exactly the three contracted kinds.
`lesson_section` is absent by
a deliberate, test-pinned refusal rather than by omission, and
advertising it would itself be
a C4.1.1 failure. Exactly one zero-entry kind remains
(`diagram`), which C4.1.1 explicitly
tolerates as pre-existing while forbidding this feature from
ADDING more — and this feature
adds none. G-KG-11 verified by `go test -count=1 -run
'TestT057|TestT058|TestGateKG11'
./pkg/search/...` -> **ok, rc 0** (in_process); pinning
assertions at `pkg/search/knowledge_search_test.go`
and `pkg/search/question_search_test.go`. Same honest boundary
as T057: the contract
amendment is uncommitted]**
````

- [x] T059 [US3] [REVIEW] Settle **U1** before implementing offsets: does the full-text engine expose match positions through a supported interface, or must offsets be recomputed by re-locating query terms in the matched text? Three-valued exit. This has a correctness edge — a recomputation can disagree with what the index actually matched, particularly under the prefix matching the lexical leg applies to the final query token. **T060 is blocked on this** (FR-029)

- [x]
````

T060 [US3] [TDD] Implement match offsets per **C4.1.3**: the lexical leg produces real offsets, **or** the field is removed and its absence stated. Leaving a field that promises a position and never carries one is forbidden. Gate **G-KG-5**. **Paired mutation**: return an empty offset list from a leg that could produce offsets (FR-029)

```
**`[BLOCKED: U1 / T059]` STRIPPED 2026-09-02 — the blocker was discharged, and the marker was
describing a state that no longer existed.** T059 settled U1 from evidence: the full-text
engine exposes match positions through a supported interface, so offsets are read from the
index rather than recomputed by re-locating query terms — the correctness edge U1 named does
not arise. Recorded at `workshop/docs/session-evidence/phase5-report.md`; the implementation
and its tests are at `workshop/platform/backend/pkg/search/offsets.go`,
`offsets_test.go` and `lexical_offsets_test.go`. Re-derive by reading those paths, not this
sentence.
```

- [x] T061 [US3] [TDD] Implement **C4.1.4**: the semantic leg reports that it **cannot** produce offsets — a different fact from producing none. One empty list meaning both is exactly the collapse this contract exists to prevent (FR-029, FR-033)

- [x] T062 [US3] [TDD] Implement the locus on every hit per **C4.1.2**, and withhold any hit whose locus does not resolve (FR-030) (FR-029, FR-030)

- [x] T063 [US3] [TDD] Prove **SC-014** over **every** hit of the benchmark run — a hit without a resolving locus fails the run, not merely itself (FR-030, SC-014)

```
**DONE 2026-09-03. The note that stood here is WITHDRAWN, not restated, and its withdrawn
claim is named so a brief written from the old text is recognisable:** *"The 'over **every**
hit of the benchmark run' half is still not implemented: `grep -c locus
pipeline/benchmark/run_retrieval_benchmark.py` returns **0**, so no hit can fail the RUN."*
That sentence remains TRUE of that Python script and it is no longer the reason this task is
open — a different instrument, built the same day, discharges
```

the task. The script is the
passage-level SC-015 benchmark; the run SC-014 names is T066's new-kinds benchmark, and it is
now executed by a gate.

Re-derived by execution, not read:

```bash
bash workshop/platform/gates/verify-retrieval-benchmark.sh   # rc=1 (served), live against :8087
#   PASS  B1 all 560 hit(s) of this run carry a resolving locus (C4.1.2 / FR-030) (served)
#   FAIL  B2 SC-015 NOT met: top-5 8/22 — a MEASUREMENT, and T067's problem, not B1's
#   PASS  B3 no negative query returned an area, term or question record
bash workshop/platform/gates/prove-retrieval-benchmark.sh    # rc=0 (population: unstated) — 6 mutations, 6 caught
```

**B1 asserts the property over EVERY hit the run returned — 560 of them, not a sample — and a
violation fails the RUN, which is the half the old note said was missing.** That is proven
rather than asserted: the paired proof's **M1-locus** mutation plants a hit whose locus does
not resolve and requires the whole gate to exit **1**; it does. The gate is three-valued by
demonstration too — **M4-no-server** and **M5-no-bench** each require exit **2**, so an
unreachable service is never a pass. Registered as `T066-T067-retrieval-benchmark` in
`platform/gates/check-registry-002.tsv`.

Honest boundary (§11.4.6), and it does not withhold the tick. The proven population is the
560 hits (served) of the new-kinds benchmark run. `pipeline/benchmark/run_retrieval_benchmark.py` still
carries no locus handling, so the passage-level benchmark asserts nothing about locus; and
`prove-retrieval-benchmark.sh` is itself **not registered** in `check-registry-002.tsv` — the
gate's row is 4 fields with no paired-proof column, so the proof runs green but nothing
enumerates it. Both are recorded rather than netted away.

■

T064 [US3] [TDD] Implement kind and area filters per **C4.1.5**, echoing the applied filter so a client can tell a filtered empty result from an unfiltered one (FR-032)

- [x] T065 [US3] [TDD] Add area and term names to suggestions per **C4.2.1**, and prove **C4.2.2**: the suggestion service still holds **no embedder**. Type-ahead stays lexical — semantic embedding under load has been observed in the tens of seconds against a 200 ms budget (FR-031, SC-016)

- [x] T066 [US3] Build the ≥20-query benchmark for the new kinds in the platform’s existing benchmark directory, beside the gates that read it, with expected identifiers fixed **in advance** (SC-015)

````
**DONE 2026-09-03. Three claims of the note that stood here are WITHDRAWN BY NAME, not
deleted, because a brief written from the old text would still assert them:**

- **WITHDRAWN claim 1:** *"it is NOT in the platform's existing benchmark directory
  (`platform/backend/testdata/benchmark/` still holds only the specs-001 `questions.tsv`)."*
- **WITHDRAWN claim 2:** *"**no gate reads it** … nothing executes the benchmark from the
  platform side."*
- **WITHDRAWN claim 3:** *"`check-registry-002.tsv` has no row naming it."*

All three were true of `pipeline/benchmark/retrieval_benchmark.json`, which is a DIFFERENT
artifact and stays where it is. A second benchmark — the one this task actually names — now
exists. Measured, not read:

```bash
ls workshop/platform/backend/testdata/benchmark/
#   kinds_retrieval_benchmark.json   questions.tsv   # (source)
python3 -c '...'   # 22 positive (area 5 · term 12 · question 5) + 6 negative (source)
grep -n kinds_retrieval_benchmark workshop/platform/gates/verify-retrieval-benchmark.sh
#   150: reads it as the gate's own input
grep -n T066 workshop/platform/gates/check-registry-002.tsv
#   62: check  T066-T067-retrieval-benchmark  shell  platform/gates/verify-retrieval-benchmark.sh
```

**Every clause of this task is satisfied by measurement:** ≥20 queries (22 positive over the
three kinds this feature indexes, plus 6 negatives); in the platform's existing benchmark
````

directory; beside the gate that reads it; and expectations fixed **in advance** — the file's own `_meta.expectations_fixed_in_advance` records that every expected identifier was taken from the CATALOGUE endpoints (`/api/areas`, `/api/terms`, `/api/areas/{id}/questions`) and **never from a search ranking**, with three stated selection rules applied verbatim and no per-query adjustment. It also records the discipline that makes that checkable: *"the file was written, then run. A query that fails is recorded as a failure, not rewritten."* The run bears that out — 14 of 22 positives (served) miss the top five and are printed as failures rather than re-cut.

Honest boundary (§11.4.6), and it does not withhold the tick: the benchmark's own RESULT is that SC-015 is not met (top-5 8/22, served). That is T067's finding, not a defect in this artifact; a benchmark whose queries all passed on first run would be the suspicious one.

☐

T067 [US3] [TDD] Prove **SC-015**: the gate prints **per-query** outcomes, not only the — **BLOCKER: retrieval quality, and nobody's decision** — the instrument is built and correct; SC-015 is MEASURED NOT MET at top-5 12/22 (54.5%), top-1 3/22, against a bar of 20/22. The `term` leg is the shortfall · **OWNER: implementer** — unblocked, but it is a quality problem, not a wiring problem aggregate — which queries fail is the useful information, and an aggregate hides it (SC-015)

PARTIAL — and the FIRST of the two reasons is DISCHARGED. The claim *"It is still **not a gate: its only exit is `return 0` (line 309) … and it still has no row in `platform/gates/check-registry-002.tsv`"* is WITHDRAWN, not restated.** It described `run_retrieval_benchmark.py`, and it is still true OF THAT SCRIPT. It is no longer the state of this task: a real gate over the new-kinds benchmark exists, is three-valued, is registered, and has a paired proof. Measured by execution 2026-09-03:

```bash
bash workshop/platform/gates/verify-retrieval-benchmark.sh   # rc=1
bash workshop/platform/gates/prove-retrieval-benchmark.sh    # rc=0 — 6 mutations, 6 caught
```

```
#   M0 control 0 · M1-locus 1 · M2-ranking 1 · M3-negative 1 · M4-no-server 2 · M5-no-bench 2
```

It prints per-query outcomes, which is precisely what this task's own descriptive clause asks for — a `id / type / kind / rank / top1 / top5 / status` row for each of the 22 positives and a `status` + `kg_records_returned` row for each of the 6 negatives, with the aggregate printed *after* them rather than instead of them. Registered as `T066-T067-retrieval-benchmark` at `check-registry-002.tsv:62`.

The SECOND reason stands and is why the box is still `[ ]`: SC-015 is measured NOT MET. The criterion needs ≥90% of ≥20 queries returning the expected item in the top five. Measured on the served path 2026-09-03 over the new-kinds benchmark: **top-5 8/22 (36.4%)** against a bar of 20/22, **top-1 4/22**. Per kind, printed by the gate: `question` 5/5 top-5 (4 of them top-1), `area` 3/5, `term` **0/12**. A `[TDD]` task that says *"Prove SC-015"* cannot be ticked while the criterion it names is red — the instrument is built, the property is absent, and the gate is correctly failing rather than passing.

Two earlier figures are SUPERSEDED, not wrong when written, and they are NOT comparable to the 8/22 above: *"top-5 20/26, top-1 8/26"* and *"13/26 before it"* were measured over `pipeline/benchmark/retrieval_benchmark.json`, whose expected identifiers are transcript and `doc_section` PASSAGES. That benchmark asks *"can the corpus be found"*; this one asks *"can the new kinds be found"*, which is the question T066 names. Quoting one as the other's before-figure would compare two different populations.

What the failure actually is, measured rather than inferred: the `term` leg returns nothing in the top five for any of its 12 queries (7 of 12 rank outside the top 20 entirely; the other 5 rank 13–20). `area` and `question` both clear the bar on their own. Recorded here because "SC-015 is 36%" reads as a uniform shortfall and it is not one.

T068 [US3] [TDD] Prove **SC-016 / SC-017**: re-run the latency harness and **publish before and** — BLOCKER:** **NOT "just a quiet host" — that remedy is now IN DOUBT, and this is the correction that matters.** Re-run 2026-09-04: rc **2**, `PASS 0 FAIL 0 UNDET 7`. The gate read the index block cleanly (generation 68, 12,979 passages, live) and then **every** endpoint refused — including the `health CONTROL`, which by the harness's own design makes the fused figure uninterpretable. Immediately afterwards `podman ps` showed `workshop-curriculum_platform_1  Up 46 seconds (starting)` and `/api/health` returned `HTTP=000`: **the container went down underneath a read-only run.** `tasks.md` already records the same sequence on 2026-09-03, so counting this it has happened **three times across two dates, every time during or immediately after `verify-search-latency.sh`**. Host load was FALLING across this run (17.5 -> 9.7), which weakens the load-average explanation the old note leans on. No causal claim is made — co-occurrence is not causation and this was not isolated — but the narrow, checkable claim is: the stated remedy *"one clean re-run, not new code"* has now failed three times identically, so treating this as bad luck risks a fourth identical result · **OWNER: implementer first** — find why the container dies under this harness's concurrent request pattern; that outranks the latency figure. **operator** second, for a quiet host, once the stability question is settled after together**. A single after-figure cannot show a regression that stayed inside the threshold (SC-016, SC-017)

```
**PARTIAL — the note that stood here is STALE and its two claims are WITHDRAWN BY NAME. The
task still does not tick, and the reason has CHANGED from "nobody re-ran it" to "the re-run
cannot be reproduced today".**

- **WITHDRAWN claim 1:** *"The newest column is still the 17:57 UTC one taken against
  **1,101** passages."*
- **WITHDRAWN claim 2:** *"No re-run against today's grown corpus is recorded anywhere."*

Both were true when written and neither is true now. `docs/limits.md` §2 carries a **fourth**
column — 2026-09-03, **2,478**-passage corpus, generation 67 — placed BESIDE the three older
ones rather than replacing them, with the breaching figure struck through and annotated
`SUPERSEDED — see §2.1` instead of deleted. That is exactly the
publish-before-and-after-together discipline this task asks for, and §2.1 states the
supersession by name. The recorded column reads `/api/suggest` p95 **9.5 ms** (SC-016's budget
```

is 200 ms) and `/api/search` fused p95 **238.6 ms** (SC-017's budget is 2 s), taken with `platform/gates/verify-search-latency.sh`, which now has a paired proof (`prove-search-latency.sh`) and two registry rows added the same day (`SC-006-search-latency`, `SC-006-search-latency-proof`).

Why the box stays `[ ]`: this reconciliation could NOT reproduce that column, and a figure nobody can re-derive is not a proof. The gate's own closing instruction is *"Re-run it; do not quote it."* It was re-run twice on 2026-09-03 and neither run is usable:

```
run 1  rc=1  search fused p95 4994.5 ms   PASS 0  FAIL 1  UNDET 4
       — but `health CONTROL` is itself UNDET (connection refused), and the container
         reported `Up 39 seconds` immediately afterwards: it went DOWN mid-run.
run 2  rc=0-with-nothing-measured        PASS 0  FAIL 0  UNDET 7
       — every endpoint including the control: "Remote end closed connection without
         response" / "Connection refused".
```

Neither run refutes the 238.6 ms column and neither confirms it — this is a COULD NOT DETERMINE, and a 2 is never a pass. The control row is the whole point of the harness: with `health CONTROL` undetermined, the fused figure is uninterpretable by the gate's own design. The host was at load average **20.2–22.3** during both attempts (§2's own caveat is that no latency figure here may be quoted without its load context) and the platform container was restarted underneath them by concurrent work in this tree.

What remains for this task is one clean re-run, not new code: `bash workshop/platform/gates/verify-search-latency.sh` on a quiet host, with the `health CONTROL` row determined, confirming SC-016 and SC-017 against the published pair. Also unresolved and recorded rather than netted away: the gate labels its budget `SC-006`, which is **spec 001's** criterion; SC-016/SC-017 are **this** spec's, and no instrument in this tree asserts them under their own names (standing rule 7).

▣

T069 [US3] [P] [SUBAGENT] Extend the search view for the new kinds, the filters and the locus display (FR-029, FR-032, FR-033)

```
**DONE 2026-09-03. The note that stood here is WITHDRAWN, not restated, and its claim is named
because it is exactly the kind a brief would carry forward:** *"The **filters and the locus
display are still absent**: `core/api.ts:273` is still `search(q: string, limit = 20)` setting
only `q` and `limit` … and the component still contains **zero** `locus` and **zero** `filter`
references."* Every clause of that was true when written and none of it is true now — the work
landed in workshop `bd7da41`.

Re-derived by reading the tree, then **proven by execution**:

```bash
grep -n 'search(q' workshop/platform/frontend/src/app/core/api.ts
#   search(q: string, opts: SearchOptions = {})   — sends `kinds` (ONE comma-joined
#   parameter, the shape the deployed server parses) and `area`; an empty filter is
#   OMITTED rather than sent empty, because "no kind filter" and "a kind filter naming
#   nothing" are different requests
grep -c 'locus'  .../features/search/search.component.ts   # 11 (source)
grep -c 'filter' .../features/search/search.component.ts   # 42 (source)

cd workshop/platform/frontend
CHROME_BIN=/usr/bin/chromium npm run test:unit
#   Chrome Headless 147: Executed 97 of 97 SUCCESS (in_process)
```

Line numbers are deliberately omitted — the file moved twice on the day — so re-derive with
`grep -n` rather than trusting a citation.

The 97 include a **`SearchComponent — T069 filters and locus`** block of 6 tests, and the six
are the task's three halves checked one at a time: no `kinds`/`area` parameter is sent when
nothing is filtered; the ticked kinds go out comma-joined with the chosen area; the kind list
is offered **from `corpus.indexed_kinds`** and never from a hardcoded list (so a kind the
corpus stops advertising disappears from the UI by
```

construction); the area list carries only
areas with an authored title and **counts the rest**; each
hit's locus is rendered **from the
server's own `locus.unit`/`locus.position`, never reassembled
client-side**; and a FILTERED
`no_match` says so where an UNFILTERED one does not, with the
echo carried onto `unavailable`
alongside a statement that the filter is not the cause.

**Honest boundary (§11.4.6), and it does not withhold the
tick:** this is a `[P] [SUBAGENT]`
task, not `[TDD]`, so no paired mutation is owed by the global
constraint — and none of the
six T069 tests is one. The evidence is 6 behavioural unit tests
executing the component, not a
grep.

▣

T070 [US3] [P] Update the route manifest and contract for the changed search and suggest endpoints (FR-059)

**CORRECTED 2026-09-03 — the note that stood here has gone
STALE, and its two claims are
WITHDRAWN BY NAME rather than deleted, so a brief written from
the old text is recognisable.**

- **WITHDRAWN claim 1:** *"all three rows still cite the
001 contract sections
  (3.6 / 3.7 / 3.11)."*
- **WITHDRAWN claim 2:** *"There are still no rows or notes for
the 002 delta sections
  §4.1 (locus/offsets/kinds/filters), **§4.2** (areas and
terms in suggest) or **§4.4**
  (progress over areas)."*

Both were true when written. Neither is true now. The one
clause of that note that still
holds is its first: `platform/gates/route-manifest.tsv` carries
`/api/suggest`, `/api/search`
and `/api/progress` rows.

**What is true, measured 2026-09-03 (source) in `workshop/
platform/gates/route-manifest.tsv`.** All
three rows now carry the **R1b compound** contract token naming
both documents —
`/api/suggest` reads `3.6+002.4.2`, `/api/search` reads
`3.7+002.4.1`, `/api/progress` reads
`3.11+002.4.4` — and each row's note names the delta clauses it
answers for (C4.2.1/C4.2.2;
C4.1.1–C4.1.5; C4.4.1/C4.4.2) alongside the tasks that changed
it, this one among them. The

manifest half of this task is therefore met for the endpoints it names.

**The COULD NOT DETERMINE recorded here on 2026-09-03 is now RESOLVED — measured the same
day, and it resolves to NOT MET rather than to a pass.** The withdrawn sentence was: *"whether
anything further is owed by the 'and contract' half of this task's title was **NOT
measured** here."* It has now been measured, and something is owed.

`contracts/http-api-delta.md` carries `## 4. Endpoints changed` with `### 4.1 GET /api/search`
(C4.1.1–C4.1.6) and `### 4.2 GET /api/suggest` (C4.2.1, C4.2.2) — so the sections exist and
the manifest rows cite them. **But §4.1's C4.1.1 still reads *"`area`, `term`,
`lesson_section` and `question` join the existing kinds"* and declares an advertised kind with
zero indexed entries a failure**, while the served corpus block advertises **seven** kinds (served)
without `lesson_section`, and `TestT057_LessonSectionIsNotAdvertisedBecauseNothingIndexesIt`
pins that refusal in code. The contract therefore requires a fourth kind the implementation
deliberately refuses, and the refusal is recorded in `tasks.md` (T057/T058) and in
`docs/limits.md` — **everywhere except the contract this task is responsible for updating.**

That divergence is the remaining work, and it is NOT a coding change: closing it needs either
a minted `lesson_section` identity or an amendment narrowing C4.1.1 to the three kinds that
can carry one — **the same spec decision T057 and T058 are already waiting on**, which is why
this task is now bound to them rather than independently open.

Separately, and outside this task: `contracts/http-api-delta.md` §2 R1b records an
open obligation on the **fourth** changed endpoint, `/api/ask`, which belongs to `002:T143`.
Its own honest-boundary paragraph says the `/api/ask` rows carry a bare `3.10`; measured the
same day, the `GET /api/ask?q=ping` row reads `3.10+002.4.3`, so that paragraph is partly
stale too and is not restated as current here.

**[TICKED 2026-09-04. The note's clause *"§4.1's C4.1.1 STILL READS `area`, `term`,
`lesson_section` and `question` join the existing kinds"* is

```
FALSE and is WITHDRAWN BY NAME**
— C4.1.1 reads THREE kinds as of the 2026-09-03 operator
amendment, so the contract divergence
that was this task's stated residue is CLOSED by the amendment
rather than by an edit here.
The route-manifest half is met: rows carry the R1b compound
contract token naming BOTH
documents, verified in `platform/gates/route-manifest.tsv`
(source) — `/api/suggest` `3.6+002.4.2`,
`/api/search` `3.7+002.4.1`, `/api/progress` `3.11+002.4.4`, `/
api/ask` `3.10+002.4.3`. The
`/api/search` row records the measured live kind list and
states the `lesson_section` refusal
in the same breath, so the manifest and the contract now agree.
Same honest boundary as T057:
the contract amendment is uncommitted]**
```

Checkpoint: everything is searchable and every hit says where — or honestly says it cannot.

Phase 6: User Story 4 — provenanced assessment (P4)

Goal: the property the reference module does not have, at 100%.

- [x] T071 [US4] [TDD] Implement the question model in `pkg/assessment/question.go`, porting the reference's field shape and **adding the mandatory citations field**. This is the feature's one deliberate divergence and the reason it exists (FR-034, FR-035)

- [x] T072 [US4] [TDD] Implement **Q1 / G-KG-2**: a question is served **only** if it carries at least one citation and **every** citation resolves. A citation resolving to redacted, not-present or could-not-determine ⇒ **withheld**. Not served with a warning, not with the citation stripped, not with an empty list. **Paired mutation**: serve it with the citation stripped (FR-035, FR-036, SC-018)

- [x] T073 [US4] [TDD] Implement **Q2 / G-KG-3**: every long-set question cites **more than one distinct** passage. **Paired mutation**: admit a single-citation question to the long set (FR-034a, SC-017a)

- [x]

T074 [US4] [TDD] Implement **Q3**: every question names the lesson sections it assesses — this is what makes per-section coverage measurable rather than estimated (FR-034b, SC-017b)

- ▣ T075 [US4] Author the short and long question sets **from the workshop's own passages** (Q5). **Nothing from the reference's 785 items may appear**, including as a placeholder (FR-004, FR-034a, FR-035)

- ▣ T076 [US4] [TDD] Implement **G-KG-16**: plant one reference question in a workshop bank and assert the boundary check fails. **Paired mutation**: scope the check to the outbound direction only. It runs both ways deliberately — outbound keeps a private recording out of public repositories, inbound keeps the workshop from shipping someone else's curriculum (FR-004, FR-004a, SC-029a)

- ▣ T077 [US4] [TDD] Implement the assembled-answer marking per **Q 4 / A3.5.4** — a client must not have to infer it (FR-038, SC-020)

- ▣ T078 [US4] [TDD] Implement `GET /api/areas/{area}/questions` per §3.5, including **A3.5.5**: report how many questions were withheld and why. A silently shorter set is how a provenance guarantee decays into a claim without anyone noticing (FR-036, FR-059)

- ▣ T079 [US4] [TDD] Implement `GET /api/areas/{area}/coverage` per §3.6: **per-area figures, never only an aggregate**; sections with zero questions listed explicitly; **no threshold and no pass field**. Gate **G-KG-12**. **Paired mutation**: replace the per-area table with a mean (FR-034b, FR-059, SC-017b, SC-017c)

```
**DONE — the "PARTIAL … the §3.6 endpoint is still not built … returns 404 … there is no
`route-manifest.tsv` row" note this task carried is WITHDRAWN, not restated.** The route was
mounted and the manifest row added on 2026-09-03; the note described a state that no longer
exists. Re-measured live 2026-09-03 against the running container
(`workshop-curriculum_platform_1`, healthy, `/api/health` 200):

```bash
# every area id from /api/areas, probed one by one
```
```

curl -so /dev/null -w '%{http_code}' http://127.0.0.1:8087/api/areas/$ID/coverage
200 for 495 of 495 areas — 0 are 404 (served)
```

The 404 half is gone and the shape holds on the WIRE, not merely on the struct. The live
response carries `per_area` as a **list**, `section_question_counts` with every roster section
as a key, `zero_sections` naming the zero-question ones again, and a `derivation` block whose
own words are `"threshold": "NONE"` and `"aggregate": "NONE"` — no threshold, no target, no
mean, no pass field. `derivation.served_only` records that withheld questions contribute
nothing (G-KG-2). Manifest row `002.3.6` is registered in
`workshop/platform/gates/route-manifest.tsv` and states the same history in its own words.
Gate and paired mutation re-run uncached and green:

```bash
cd workshop/platform/backend
go test -count=1 -run 'TestT079' ./pkg/assessment/... ./internal/api/...   # ok (in_process)
# pkg/assessment/coverage_test.go:154 TestT079_PairedMutation_ReplacePerAreaTableWithMean
# internal/api/coverage_wire_test.go — 7 wire tests, incl.
#   TestT079_Wire_ExpressesNoThresholdAndNoPassField and
#   TestT079_Wire_ForbiddenKeyScannerActuallyCatchesOne (the scanner's own paired proof)
```

Honest boundary (§11.4.6): A3.2.2's publication-review requirement is deliberately **not**
applied on this route, by a call whose reasoning is recorded in `internal/api/coverage.go`'s
own doc comment — which is why coverage answers 200 for all 495 areas (served) while `GET
/api/areas/{area}` answers 200 for only **2**. §3.9 makes the opposite call; both divergences
are declared where they happen rather than left to be discovered. Separately, the live
`derivation.lesson_section_identity` field warns that the two roster sizes it reports do not
measure the same thing — read it before comparing them.

- [x] T080 [US4] [TDD] Prove **SC-018** by enumerating every served question and resolving every citation, and **record the reference module's measured value — 0 of 785 — beside it**, so the comparison is measured rather than claimed (FR-035, SC-018)
```

- [x] T081 [US4] [TDD] Prove **SC-019**: the question-to-moment-and-back round trip, over every eligible question; if impractical, a seeded sample of ≥30 **with the population size published** (FR-037, SC-019)

- [x] T082 [US4] [TDD] Extend progress to areas and question sets, and prove **C4.4.2**: progress **survives a content re-ingest**. The reference deliberately preserves its progress table while wiping everything else; a learner's history is not a derived artifact (FR-039)

- [x] T083 [US4] [P] Add route-manifest rows and contract sections for the assessment endpoints (FR-059)

````
**DONE — the "PARTIAL … coverage (§3.6) and export (§3.9) still have neither a manifest row
nor a mounted route … both return 404" note is WITHDRAWN, not restated.** All three assessment
endpoints now have a contract section, a manifest row and a mounted route. Re-measured
2026-09-03:

```bash
grep -n '^### 3\.' specs/002-knowledge-areas-deep-linking/contracts/http-api-delta.md
# 3.5 questions · 3.6 coverage · 3.9 export — all present (source)
grep -c '002\.3\.[569]' workshop/platform/gates/route-manifest.tsv   # 3 rows (source)
curl -so /dev/null -w '%{http_code}' http://127.0.0.1:8087/api/areas/$ID/export   # 200 (served)
```

Manifest rows `002.3.5`, `002.3.6` and `002.3.9` are registered, and the two new rows record
in their own words that they "WAS A BARE 404 UNTIL 2026-09-03" — the row and the route landed
together, which is the pairing this task exists to enforce. Export behaves per contract on both
sides of the gate, measured live (served) rather than read:

- **Published area** (`01M1GWW49GNKYBEXFFCNRWM1SY`) → `exportable: true`, four formats listed,
  `markdown` `present` with a real `href` and `size_bytes`, `html`/`docx`/`pdf`
  `could_not_determine` with `toolchain_blockers` naming `pandoc`/`weasyprint` as absent **in
  the container** — A3.9.2's "never an empty format list"
````

```
honoured.
- **Unpublished area** (`01M1GWQYWSKZA7A2S59MNV0Z2Q`) → `exportable: false`, four
  `precondition_blocked` formats, no `href`, no source path. A3.9.1 requires exporting nothing
  **and saying so**; saying so needs a body, so this is a 200 carrying the failed precondition
  rather than a 404. The §3.2 route keeps returning its own `404 area_not_published` — this
  does not soften it.

```bash
cd workshop/platform/backend
go test -count=1 -run 'TestT083' ./internal/api/...   # ok (in_process)
# 10 wire tests, incl. TestT083_ProbeExportToolchain_NeverAsksForAVersion —
# the probe makes each tool DO its job rather than answer --version
```
```

- [ ] T084 [US4] [REVIEW] Review question provenance end to end before the practice surface consumes — **BLOCKER:** none but the work — the only review record is `phase6-report.md` (unchanged since 2026-09-02 12:54) and it merely mentions T084; no later record exists. Pure review-record authorship, and the second of the two things holding T122 · **OWNER: implementer/reviewer** — unblocked today it (FR-035, SC-018)

```
**NOT DONE (re-measured 2026-09-03, unchanged).** `workshop/docs/session-evidence/phase6-report.md`
deferred this review pending T075 and T078. **Both have since been completed and the review
was never revisited** — `phase6-report.md` is still the newest phase-6 artifact (2026-09-02
12:54), and `grep -rl T084` across `docs/` returns only that report and its own brief. No
later review record exists under `docs/session-evidence/`, `curriculum/` or `specs/002-*/`.
```

Checkpoint: every served question is traceable to the moment that teaches it.

Phase 7: User Story 5 — the learning surface (P5)

Goal: a learning platform, with its own identity, that loses nothing it already had.

- [x] T085 [US5] **Enumerate the four existing capabilities — chapter list, transcript, recording player, cross-references — and write the list down BEFORE any reshaping**. SC-023 is a before-and-after comparison, and a list reconstructed afterwards is a recollection, not a measurement (FR-041, SC-023)

- [x] T086 [US5] [P] [SUBAGENT] Build the area list and area detail views (FR-040)

- [x] T087 [US5] [P] [SUBAGENT] Build the practice view with the question-to-moment jump and return (FR-037, FR-040)

- [x] T088 [US5] [P] [SUBAGENT] Build the progress view across areas (FR-039, FR-040)

- [x] T089 [US5] [P] [SUBAGENT] Build the study-plan view (FR-040)

- [x] T090 [US5] Subordinate the existing media views under areas per **D 2** — reachable from within an area rather than as top-level objects. They are the working half of what exists and are what makes a timestamp jump possible at all (FR-041, SC-023)

- [x] T091 [US5] [TDD] Prove **SC-023** against T085's written list: every enumerated capability is still reachable (FR-041, SC-023)

- [x] T092 [US5] Give the workshop its own token **values** against the shared contract. **Do not fork the component library** — forking doubles maintenance and guarantees drift (FR-043) (FR-042, FR-043)

- [x]

T093 [US5] [TDD] Prove **SC-022** with **both** checks: the design toolkit's token conformance check **and** a literal-value scan. Either alone passes a stylesheet that defines perfect tokens and ignores them (FR-042, SC-022)

- [x] T094 [US5] Implement interaction and motion from the platform's ranked, sourced user-experience research, each decision naming the finding it rests on (FR-044a). An effect with no supporting finding does not ship on the grounds that it looks better (FR-044a, SC-029b)

```
**`[BLOCKED: UX research]` STRIPPED 2026-09-02 — the blocker was discharged, and the marker was
describing a state that no longer existed.** The research this task waits on exists as a real
artifact: `workshop/docs/research/ux-research-2026-09-01.md`, 1,231 lines (source), carrying a **ranked**
proposals section (16 proposals, ordered by value per unit of effort with stated effort bands),
a separate rejected-with-reasons section, and 194 source URLs — ranked and sourced, which is
exactly what FR-044a requires it to be. `workshop/docs/session-evidence/phase7-report.md`
records FR-044a/b (T094/T095) as citing the finding each effect rests on. Re-derive by reading
those paths, not this sentence.
```

- [x] T095 [US5] [TDD] Implement reduced-motion support: final visual state preserved without the transition, and no effect load-bearing for comprehension (FR-044b) (FR-044b)

- [x] T096 [US5] [TDD] Prove **SC-021**: an automated audit per view plus a keyboard-only traversal that **visits every deep-link affordance** and asserts each can be reached, activated and returned from (FR-044, SC-021)

Checkpoint: the product is a learning surface and nothing was lost getting there.

Phase 8: User Story 6 — four-format export (P6)

- [x] T097 [US6] Port the export toolchain design as-is, driving it from the **capability probe** in T004 rather than from a name on `PATH` (FR-003, FR-045, FR-046)

- [x] T098 [US6] [TDD] Implement **G-KG-9**: an unavailable toolchain yields **could not determine**, never an empty format list (FR-049). **Paired mutation**: return an empty list instead (FR-049, FR-054)

- [x] T099 [US6] [TDD] Implement the publication precondition: an area without a review exports nothing **and says so** (FR-048, A3.9.1) (FR-048, SC-007)

- [x] T100 [US6] [TDD] Implement citation preservation across all four formats (FR-047) (FR-047)

- [ ] T101 [US6] Port diagram rendering from text source — reproducible, unlike an embedded binary — **BLOCKER: operator decision** — `mmdc -V` exits 0 but an actual 2-node render exits **1 with no SVG**, `grep -c mermaid` is **0** across all 5 area documents, and there is no decision on where a diagram lives in the seven-section skeleton · **OWNER: operator** (content/contract placement), then **implementer** (FR-014a) (FR-014a)

```
**OPEN. Re-measured 2026-09-03, and leaving it UNWIRED is now backed by three separate
disqualifying findings — each one on its own is enough to rule out "make `export_area` call
`render_diagram`" as the fix.** The renderer function is implemented and covered in isolation;
the function that assembles a published area's four formats does not invoke it. Re-derived
here rather than read:

```bash
grep -rn 'render_diagram(' workshop/pipeline/ --include=*.py | grep -v 'def render_diagram'
# 5 call sites, all in the two test/verify modules; export_area is not among them
```

1. **The renderer is unusable, and asking it for a version hides that.** Measured directly on
   this host, not carried forward:
```

   ```bash
   mmdc -V                                  # 11.12.0   rc=0
   printf 'graph TD\n A-->B\n' > t.mmd
   mmdc -i t.mmd -o t.svg; echo "rc=$?"     # rc=1 — and no t.svg is written
   ```

   The version query succeeds; the actual 2-node render fails and produces no file. This is the same trap the export contract's own **A3.9.2** warns about — a name on `PATH` and a version string are not evidence of capability, so capability has to be exercised. Wiring the call on the strength of the version query would break every published area instead. Worth naming: T083's export probe refuses to accept a `--version` answer for exactly this reason, and this task is the case that shows the refusal earning its keep.
2. **There is no input to render.** `grep -c 'mermaid' workshop/docs/training/areas/*.md` returns **0** for each of the five area documents, so the new code path would be dead for every document that exists.
3. **Nobody has decided where a diagram lives in an area document**, and this is the binding ground. FR-014a requires text-source authoring but is silent on placement, and the SC-005 skeleton fixes seven headings with none for a diagram. Choosing a heading name, a fence tag, and whether an area may hold more than one is a content/ contract decision, not a coding one.

A justification that overstated its own support has been corrected upstream, and it is logged here instead of being absorbed without trace. Eighteen Mermaid sources across six files sit under `workshop/docs/training/diagrams/` — held outside the area documents precisely so they do not pre-empt finding 3. The README there defended that placement partly on the grounds that the platform's own front end displays them. **It does not**, measured 2026-09-03: `grep -rn mermaid platform/frontend/src/` returns **0**, and none of the Angular workspace's 23 dependencies is a Mermaid renderer. The README has been fixed inside `workshop/`. The fences do display in GitHub/GitLab viewers, which is what makes source form a defensible choice — but not here.

Full investigation, both measurement dates, and the defects row `render-diagram-never-called`: **`workshop/docs/limits.md` §10.5**.

- [x] T102 [US6] [TDD] Prove **SC-024**: the file-existence matrix must be **complete**. Record the reference's measured 25-of-34 beside it, so nobody later matches the reference and calls it done (FR-045, SC-024)

DONE. Every clause of the PARTIAL note this task carried is WITHDRAWN, not restated — it
said the §3.9 route still 404s, that `export.py` hardcoded `review=None`, that the matrix had
never been exercised over real content, and that the reference's 25-of-34 was not recorded
beside a workshop matrix. All four were true when measured earlier on 2026-09-03 and **all four
were fixed later the same day**, while this reconciliation was in progress. Re-measured after
the change, by execution:

```bash
cd workshop/pipeline/extract
../venv/bin/python -c "import export; print(export.real_materials_export_status(produce=False))"
# rc=0 (in_process) — 2 published area(s) of 5; 3 blocked by A3.9.1; 0 blocked by SC-012
#        matrix: 2/2 area(s) carry all four formats
bash workshop/platform/gates/verify-sc024-export-matrix.sh    # rc=0 (served), live against :8087
```

Both halves of the task are satisfied and both were watched, not assumed. The matrix is
complete — 2 of 2 (in_process) published areas carry markdown, html, docx and pdf. The reference's
25 of 34 (source) is printed on the same run, and the gate labels it in its own output as recorded
beside the result and never as the bar — 25/34 is 73.5%, and SC-024 requires 2/2, so
matching the reference would specifically NOT be this criterion met. That is the exact
misreading this task was written to prevent.

The fix is worth recording, because two defects were hiding each other. The old caller
passed `review=None` — a constant, so the branch it selected described the function rather than
the tree — and it passed the document's filesystem **path** where reviews are keyed by the
minted **ULID**, so a review that HAD loaded could never have

```
matched. Neither was observable
while the other stood: nothing looked a review up, so the key
mismatch never got the chance to
fail. Identity now resolves through the same production
promotion path the authoring stage
uses. `curriculum/publication-reviews.jsonl` holds **2**
reviews (source), and they are the same 2 area
ids — of 495 — that `GET /api/areas/{area}` serves at 200
(served), so the Python and Go paths now agree
where they previously disagreed. `real_materials_export_status`
is read-only by default;
producing artifacts is an explicit `--produce-real-materials`
step.

**Paired mutation, executed rather than cited**
(`verify_export.py --prove-failure`, rc=0, in_process): with
the review store mutated to return nothing, the function drops
to 0 published areas and rc=2 —
*the pre-fix behaviour reproduced exactly* — and unmutated
returns 2 and rc=0. Registered as
`T102-review-identity-not-hardcoded` in `platform/gates/check-
registry-002.tsv` with its proof
named in the row.

**Honest boundary (§11.4.6), and it does not withhold the
tick:** the SC-024 shell gate itself
(`verify-sc024-export-matrix.sh`, registered as `T102-sc024-
export-matrix`) carries **no**
`--prove-failure` and no paired-proof column — `bash … --prove-
failure` returns rc=2 (in_process) `unknown
argument`. The paired mutation above covers the identity
mechanism the result turns on, not the
matrix gate's own failure mode. Under this file's global
constraint that **every** check owes a
paired mutation, that shell gate still owes one.
```

■

T103 [US6] [TDD] Prove **SC-025** by **extracting text and diffing**, not by byte comparison — embedded timestamps make byte equality unachievable for some formats, and a criterion nobody can meet is worse than none (FR-046, SC-025)

Phase 9: User Story 7 — the repeatable pipeline (P7)

Goal: the next chapter runs through unchanged. This is the phase that distinguishes a finished chapter from a finished feature.

- [x] T104 [US7] Extend the platform's **existing** chapter-addition procedure and its prompt with the knowledge-layer stages. **Do not create a second way to add a chapter** (S1) — two procedures for one act guarantee one of them rots (FR-033d, FR-033e)

- [x] T105 [US7] [TDD] Implement three-valued exits and resumability on every new stage (S3, S4), with progress projected from **measure d rate**, never from an estimate (FR-033g, FR-054)

- [x] T106 [US7] [TDD] Implement the taxonomy **update** path (S6): established areas keep their identifiers and gain evidence, newly evidenced areas are added, contradictions are reported (FR-033f)

- [x] T107 [US7] [TDD] Prove **SC-015d**: capture the taxonomy before and after, assert **0** established identifiers changed and the contradiction report is non-empty when one was seeded. **Paired mutation**: re-derive area identifiers on each run (FR-033f, SC-015d)

- [x] T108 [US7] Build a **small synthetic chapter** fixture — synthetic, because it must contain no workshop content and it must exercise the **minting** path (FR-057, SC-015c)

- [ ] T109 [US7] [TDD] Prove **G-KG-17 / SC-015c**: run the whole pipeline against the synthetic — **BLOCKER:** none but the work — `prove_g_kg_17_synthetic_chapter` self-scopes in its own docstring ("SCOPED HONESTLY") to extracted areas/themes plus taxonomy, while T109 asks for EVERY S2 output · **OWNER: implementer** — unblocked today chapter; assert every output in S2 exists and the diff contains **no hand-created structural file and no code change**. **Paired mutation**: remove one stage; the gate must go red naming the missing output. **Do not substitute a re-run of the real chapter.** That tests idempotency, which is SC-004; it cannot test whether a *ne* *w* chapter works, because every identifier it needs already exists — the run would pass by matching and never by minting (FR-033d, SC-015c)

```
**PARTIAL (re-measured 2026-09-03, unchanged).** G-KG-17
exists, is registered
(`G-KG-17-synthetic-chapter`) and runs the REAL
`run_pipeline.run()` entry point against the
synthetic chapter. Its own docstring
(`prove_g_kg_17_synthetic_chapter` in
`pipeline/extract/verify.py`, "SCOPED HONESTLY") still scopes
it to extracted areas and themes
plus the taxonomy file, and names the remainder of S2's output
set — authored materials,
question sets, cross-references, index entries, deep links — as
owned by
`platform/backend` or by stages not yet built, and explicitly
does not claim to exercise them.
T109 asked for **every** output in S2.
```

- [x] T110 [US7] [TDD] Prove **G-KG-18** / **SC-015e**: withhold one required input; assert the run names exactly what is missing and publishes nothing. **Paired mutation**: downgrade it to a warning (FR-033g, SC-015e)

- [x] T111 [US7] [TDD] Prove **S7**: no stage writes to a source. Assert every source file's size, modification time and inode are unchanged after a full run. **Paired mutation**: have a stage rewrite a source in place (FR-?)

- [x] T112 [US7] [TDD] Prove **S8**: evidence is written for every outcome, **e specially** could not determine — the run that determined nothing is the one a reader most needs the record of. **Paired mutation**: skip evidence writing on the could-not-determine path (FR-054, FR-055)

Checkpoint: the pipeline is observed producing a complete chapter, not reported as doing so.

Phase 10: Polish and honest limits

- [x]

T113 [TDD] Extend retrieval — **not generation** — over the new kinds (C4.3.1). Safe because the retrieval gate refuses before any model runs (FR-028, FR-051)

- [x] T114 [REVIEW] Settle **U4**: does an entailment model load on this host? Three-valued exit. The platform's code claims it refuses to degrade silently; **a claim is not a measurement** (FR-052, FR-054)

- [x] T115 [TDD] Implement the **answer-against-question** verification layer (FR-051) in the existing support-verifier seam. All four current layers verify the claim against the **passage**; none verifies it against the **question**, which is why a topically related fragment that does not answer what was asked passes all four (FR-051)

````
**`[UNBUILT: decision taken 2026-09-02]` DISCHARGED 2026-09-03
— the layer was BUILT, WIRED
AND PUSHED, and the marker was describing a state that no
longer existed.** The code the
marker was waiting on exists and runs: `platform/backend/pkg/
answer/question.go` (L5, the
deterministic question-demand floor) plus a new `platform/
backend/pkg/answerhood/` (the model
answerhood judge — deliberately **not** `pkg/entail`, because
entailment relates
passage→claim and answerhood relates question→claim, and
conflating them is the defect this
task names). Wired in `pkg/answer/pipeline.go` **after L4, not
instead of it**: a
`VerdictDeclined` yields `ReasonDoesNotAnswer`, so the
topically-related fragment this task
was written about is now refused. All of it landed in workshop
`692a27a`, pushed.
Re-derived 2026-09-03, not read from prose:

```bash
bash workshop/platform/gates/verify-answer-question.sh
# rc 0 (population: unstated) — 10/10 L5 properties
bash workshop/platform/gates/prove-answer-question.sh
# rc 0 (population: unstated) — CAUGHT 3, MISSED 0
curl -s http://127.0.0.1:8087/api/ask/status | grep
question_verifier_kind
#   question-focus+llm (served) — L5 is live on this
deployment, not merely compiled
```

Registered as `T115-answer-against-question` in `platform/
gates/check-registry-002.tsv`.
**Honest boundary, and it does not withhold the tick:** `docs/
````

```
limits.md` §10.15 records the
layer's measured cost and its four remaining weaknesses —
fabrications fell 11 → 1 (population: unstated) on the
57-question benchmark and **spec 001's** fabrication criterion
is still NOT met; floor 1 is
strong on three demand classes and blunt on four; the judge
defaults to the same model that
generated the claim; L5 checks that the question was ANSWERED,
never that the answer is
CORRECT. Those are recorded tuning limits of a layer that
exists. **Spec 002's own SC-010 is
the identifier-survival criterion (`spec.md`), proven under
T053 — it is a different
criterion that happens to share a number, and it is not this
task's acceptance.**
```

- [x] T116 [TDD] Implement **C4.3.4**: a verifier that was requested and could not load reports **could not determine** and does **not** degrade to the weaker one (FR-052, FR-054)

```
**`[UNBUILT: decision taken 2026-09-02]` DISCHARGED
2026-09-03.** The build-order dependency
on T115 is discharged because T115 is built, and C4.3.4 landed
with it in workshop `692a27a`.
`pkg/answer/pipeline.go` maps a requested-but-undecidable
verifier
(`grounding.VerdictUnavailable`) to
`CodeQuestionVerificationUnavailable` and **returns**
— it does not fall through to `answered` on the strength of the
layers that did run, which is
what "does not degrade to the weaker one" means written as code
rather than as a promise.
Proven, not asserted: `prove-answer-question.sh` seeds `m2-
silent-degrade` — a requested
judge that could not decide being silently dropped — and the
gate catches it (`CAUGHT 3
MISSED 0` (population: unstated), re-run 2026-09-03).
Registered separately as
`T116-question-verifier-undetermined`. A nil verifier is a
different case and is not treated
as a pass either: `/api/ask/status` reports
`question_verifier_kind: null` with prose naming
what is therefore unchecked.
```

- [x] T117 Update the shipped limits document to name **every** open defect, including the measured fabrication rate on unanswerable questions and the undefended topically-related-but-non-answering case (FR-050) (FR-050, SC-026)

**DONE 2026-09-03. The note that stood here named its own closure condition and that condition
is now met, so the claim *"It is measurably incomplete **right now**: `bash
workshop/platform/gates/verify-limits-completeness.sh` exits **1** — `MISSING
area-term-over-generation`, now 1 of **14** registered defects"* is WITHDRAWN, not restated.**
The one-line fix it predicted was made, and the denominator moved again. Re-measured by
execution, not read:

```bash
bash workshop/platform/gates/verify-limits-completeness.sh
#   rc=0 (source) — checked 15 registered defect(s) against docs/limits.md
#   OK: all 15 registered defect(s) are named in docs/limits.md.
```

The fix was to the ANCHOR, not to the document's coverage, and it is worth recording because
the failure mode was an instrument pointing at moving text: `defects-registry.tsv`'s
`area-term-over-generation` row had quoted §10.9's heading, a retitle removed it, and the gate
went red **for a document that had merely been corrected**. Its replacement anchor carries no
moving count on purpose, and says so in its own row.

**The two items this task names explicitly are both present, checked one at a time (source) rather than
inferred from the gate's exit code:** the measured fabrication rate on unanswerable questions
is in §1.1 (2 of 10) and again in §10.15's before/after table (**11 → 1** on the 57-question
benchmark, with the corpus-dependence of that figure stated); and the undefended
topically-related-but-non-answering case is named in §10.11 in the contract's own words —
*"none of the four existing verification layers checks a claim against the QUESTION, only
against the passage, so a topically-related-but-non-answering fragment could pass all four
unnoticed"* — with §10.15 recording the L5 layer that now defends it **and its four remaining
weaknesses**.

Honest boundary (§11.4.6), and it does not withhold the tick: *"every open defect"* is
bounded by what `platform/gates/defects-registry.tsv` registers — **15** rows today (source), up from

```
13 then 14. A defect nobody registered is invisible to this gate by construction, which is the
same bound SC-026/T118 measures and not a new one. The count has moved three times in two
days; re-run the gate rather than quoting 15.
```

- [x] T118 [TDD] Prove **SC-026**: cross-check recorded defects against the limits document; an unnamed defect fails. **Paired mutation**: remove one defect from the document (FR-050, SC-026)

- [ ] T119 [TDD] Prove **SC-028**: drive every new check into a could-not-determine condition and — **BLOCKER:** none but the work — SC-028 coverage reaches only 3 Phase-10 checks; the 18 G-KG gates have never been driven to state 2, and the registry population has since grown to 70 checks · **OWNER: implementer** — unblocked today assert the **third** state, not either of the other two (FR-054, SC-028)

```
**PARTIAL (re-measured 2026-09-03, unchanged).** `platform/gates/prove-sc028-undetermined-states.sh`
exists and drives the three Phase-10 checks (T114/T118/T120) into their third state. The
**18 G-KG-1..18 gates are still not independently re-driven** into a could-not-determine
condition; `check-registry-002.tsv`'s `debt sc028-retroactive-coverage` row is still present
and still records that as a gap rather than assuming compliance. **The population also grew
on 2026-09-02** — five new rows landed (`T115-answer-against-question`,
`T116-question-verifier-undetermined` and the three `T041-*` R3 rows) — so "every new check"
now covers more than it did when this note was first written, not less.
```

- [x] T120 Register every new check in the check registry (SC-027). Note the registry prints known debt on every run **by design** — a zero exit means every check is accounted for, **not** that every check has a paired proof (FR-053, SC-027)

- [x] T121 Update the platform’s documentation set — quickstart, user guide, manual, FAQ — for the knowledge layer, and state plainly what it cannot do (FR-050)

- [ ]

T122 [REVIEW] Final content-boundary check **in both directions** (SC-029, SC-029a), plus the — **BLOCKER: predecessor T042 (SC-029a inbound), plus T022 and T084 as a final review** — 2 of its 3 halves are already green: zero workflow files, and `CONTINUATION.md` carries 19 spec-002 references · **OWNER: implementer**, last; needs T042/T022/T084 first fleet-wide no-CI gate (SC-030), plus `CONTINUATION.md` synchronised (FR-056, FR-057, SC-029, SC-029a, SC-030)

```
**PARTIAL (re-measured 2026-09-03, unchanged).** Two of three halves hold: the fleet no-CI
gate is green (`git -C workshop ls-files '.github/workflows/*'` → **0**), and the umbrella
`CONTINUATION.md` carries its spec-002 section. The **"both directions"** content-boundary
half is still not satisfiable — SC-029a is not built (`docs/limits.md` §10.14, defects row
`sc029a-not-built`), the same gap as T042. This is a FINAL review task in any case: it cannot
honestly close while T022, T041 and T084 are open above it.
```

Phase 11: On-screen text — OCR ingestion (added 2026-09-02, decision D5)

Every task in this phase is `[UNBUILT]` in the exact sense the marker legend defines: the clarification it waited on is SETTLED, the operator's decision is TAKEN, and what remains is code nobody has written. The operator chose **spoken PLUS on-screen OCR**, extending this specification **in place**, and then chose to **specify now and build later**. No checkbox in this phase may be ticked by the act of writing it down.

"Nothing here is built" is WITHDRAWN as of 2026-09-03, and the correction is deliberately small: one HALF of one task is built, and it is still unticked. `workshop/pipeline/detect_ocr.sh` now exists — the T123 toolchain capability probe, three-valued, registered in `platform/gates/check-registry-001.tsv` as `ocr-toolchain-capability`, with a `--prove-failure` that executes 6 mutations.

“Twenty of twenty boxes in this phase remain unticked” is WITHDRAWN as of 2026-09-03, and so is every sentence that rested on it. It is NINETEEN of twenty. T125 is built and ticked. The three measurements that supported the old figure are each now false and are named individually rather than quietly dropped:

- ~~`grep -rl screen_text` over `platform/`, `pipeline/` and `docs/` returns **nothing**~~ — it now returns `platform/backend/internal/passagestore/domain.go`, `platform/backend/internal/passagestore/screentext_test.go`, `platform/backend/pkg/search/catalog_test.go`, `platform/gates/check-registry-002.tsv` and `pipeline/detect_ocr.sh`.
- ~~T125’s passage kind, on which T126–T141 all sit, still does not exist~~ — it exists, `passagestore.KindScreenText`, with gate **G-OCR-2** and a 13-mutation paired proof registered in `platform/gates/check-registry-002.tsv`.
- ~~an empty `platform/backend/ocrprobe/` directory~~ — still empty, and still not load-bearing. **The kind was NOT built there.** `ocrprobe/` would have been a second home for OCR vocabulary; the kind belongs in the passage store beside the four it joins, and putting it anywhere else is the “fifth registry” mistake in miniature.

The file-level count “104 ticked / 39 unticked” is WITHDRAWN, not restated: it is 106 / 37 of 143. The intervening 105 / 38 is also withdrawn — another agent ticked one box outside this phase between the two readings, so 104 was already stale when it was written down and 105 was never recorded here at all. Exactly one of the two moves is this phase’s: T125.

Read the per-task notes below before planning any of this: each of T123–T142 carries a measured blocked-note naming what blocks it and on whom, and the three classes are not interchangeable — **blocked on an OPERATOR** (T124, T130, T132, and the missing speech-WER baseline behind T135), **blocked on a CAPABILITY that was measured rather than assumed** (the per-chapter language-detection half of T123), and **merely unwritten** (T139, which is unwritten but must not run first, and — newly, as a direct consequence of T125 landing — **T129**, whose only recorded blocker was the passage kind).

T125 landing moved four notes below, and they have been rewritten rather than left to rot. T126 is now blocked on T124 alone; T127 on T126 alone; T128 on T124 alone; and **T129 is no longer**

blocked at all. A dependency note that still names a satisfied blocker is worse than no note — it sends the next reader to wait for something that already happened.

Two preamble claims below are now measured rather than inferred, and one of them changed sign. The OCR engine is not merely present: it reproduced a known fixture exactly, returned per-word confidence and geometry, and survived a video encode → frame-sample → recognise round trip. What does NOT work is the language-detection half — tesseract OSD misclassified an all-caps Latin fixture as Cyrillic **at the highest confidence of the three fixtures tried**, so the wrong answer is the confident one. The consequence for planning is specific: a task that assumed on-screen language could be detected per chapter from the engine's own signal was assuming something this host does not provide, and the substitute is a bake-off against T132's hand-truthed sample — which makes that half of T123 downstream of an operator task, not of a decision.

Sequenced after the tasks this specification already carries unbuilt — 18 as of 2026-09-03, down from 28 when this phase was written, then 26, then 23, then 22, as T115/T116, T079/T083, T102 and T063/T066/T069/T117 were built. That is a deliberate ordering, not a formality: on-screen mentions attach to areas, feed coverage figures and are exported alongside spoken ones, so a Phase-3-to-Phase-10 defect reached through OCR-derived evidence would be diagnosed twice.

Purpose: make the substance of a screen recording addressable — and make its accuracy a measured quantity before a single on-screen mention is published.

Why the accuracy obligation is the load-bearing half of this phase. The risk in OCR is not that it fails to run; it runs on this host. The risk is that it runs **well enough to look right** and deposits thousands of passages whose text is approximately correct and whose timings are approximately placed — degrading the measured accuracy of a corpus that currently has one, while every published figure keeps looking precise. Speech recognition here is calibrated with a recorded procedure (`workshop/pipeline/CALIBRATION.md`, `workshop/pipeline/compare_engines.py`, `workshop/scripts/verify-accuracy.sh`, all three present, measured 2026-09-02). **A task that said "OCR the video" without an accuracy budget would be exactly the defect this spec's own clarification was raised to prevent.**

Two host facts, measured 2026-09-02, and one that is NOT a capability claim. The OCR engine resolves at a **user-local** path and reports **three** language packs — consistent with contract §7 V1's finding that every export tool on this host resolves user-locally, and the reason T123 probes rather than assumes. ~~**No OCR artifact of any kind exists in the workshop today** (`find workshop -iname '*ocr*'` returns nothing outside vendored environments), so every task below is genuinely unbuilt.~~ **WITHDRAWN 2026-09-03: one artifact now exists — `workshop/pipeline/detect_ocr.sh`, the T123 toolchain probe. The same `find` now returns it. Every task below is still genuinely unbuilt as a TASK, and the probe is exactly the reason that sentence has to be rewritten rather than reused: an artifact existing and a task being done are different claims.** The "three language packs" figure above also needs reading with care — re-measured 2026-09-03 they are `eng`, `rus` and `osd`, and **`osd` is a script-detection model, not a recognition language**, so this host reads **two** languages, not three. And a boundary that must not be misread: **every deep link in this deployment resolves at SEGMENT width** — `workshop/docs/limits.md` §10.1 and the defects row `precision-segment-only`. **"Joins the existing precision model" does not mean "word precision is live."**

Read that boundary from the SPAN, never from the `precision` field — the two disagree today. An attempt was made on 2026-09-03 to withdraw the sentence above on the strength of the field having flipped to `word` after commit `ff90e09`; **that withdrawal was itself wrong and has been reversed.** Measured the same day over 40 areas, entries labelled `word` have a median span of **7.650 s** against **7.620 s** for entries labelled `segment` — the same width, and the recorded segment distribution (median 6.74 s). `pkg/knowledge/wordjoin.go`'s `JoinSegmentWords` returns the first-to- last word of the **whole segment**, so the flag reports sidecar completeness, not span width. See T052's note. A `word` label on a 7-second span is a defect to fix, not a capability to build on — and Phase 11 must not assume it.

Independent test: with the OCR stage run over one chapter, open any area and confirm that its evidence distinguishes what was said from what was shown, that the two are counted once where they coincide, and that both accuracy figures are published beside the speech-recognition ones.

- [ ] T123 [UNBUILT] [P] Gate **G-OCR-1** — capability-probe the OCR toolchain per contract §7 **V1**: — **BLOCKER: half BUILT, half blocked on CAPABILITY not on a decision** — the `[UNBUILT]` marker is STALE: `pipeline/detect_ocr.sh` EXISTS (754 lines), is registered as `ocr-toolchain-capability` with a paired proof, and its rc-2 probe is verified live (`--scratch-dir /nonexistent` -> rc 2). What is NOT buildable is per-chapter language detection: tesseract OSD misclassified an all-caps Latin fixture as Cyrillic at the HIGHEST confidence of three fixtures, so a confidence floor selects FOR the error. Resolving it needs a bake-off against T132's hand-truthed sample · **OWNER: implementer** for the built half; **blocked on T132** for the language half invoke the engine with **the flag that matters**, never `--version`, and detect the recording's on-screen language **per chapter** per **V2**. Engine absent, or the needed language pack absent, yields **could not determine** — never empty text presented as "nothing on screen". **Paired mutation**: probe by name on `PATH` only; the gate must go red. This repository has already shipped a defect of exactly this shape — a media tool that answered a version query and rejected the flag that mattered (FR-053, FR-054) **PARTIAL — measured 2026-09-03. The box stays unticked because half of this task is BUILT and half is not, and the unbuilt half is blocked on a capability that was MEASURED, not assumed. BUILT**: `workshop/pipeline/detect_ocr.sh` — three-valued (0 usable / 1 unusable / 2 could-not-determine), registered in `platform/gates/check-registry-001.tsv` as `ocr-toolchain-capability`, with `--prove-failure` executing **6** mutations (M1 healthy, M2 absent, M3 version-only, M4 silent, M5 no-wordbox, M6 undetermined) and a CONTROL asserting the M3 fake still passes a naive `command -v` + `--version` probe — the exact defect shape this task names. Every tool is exercised with the flag it will be used with, against a fixture whose text is KNOWN, and the output is COMPARED to that text: "non-empty" is not accepted, because a recogniser pointed at the wrong script returns confident nonsense rather than nothing (measured: a Cyrillic fixture read with `-l eng` returned a fluent-looking Latin transliteration, rc 0). Measured on this host, exit **0** — tesseract 5.5.2 at a user-local path reproduced a 29-word fixture exactly, `tsv` returned 32 scored word rows all carrying positive geometry (mean conf 96.2), packs are `eng,rus` (`osd` is script detection and is NOT counted as a language), and an ffmpeg encode -> `-vf fps=1` sample -> OCR round trip returned the fixture exactly. **Two host traps were re-measured while building it, and both shaped the design.** The `ffprobe` symlink trap this repository already documents was reproduced independently (answers `-version` rc 0, rejects `-show_format`); and a second one that was not on record: the same ffmpeg advertises `--enable-libfreetype --enable-fontconfig` in its own `-version` configuration string and has **no `drawtext` filter**. A tool's self-description is not a capability measurement either, which is why the probe fixture is EMBEDDED rather than drawn at run time. **NOT BUILT — the per-chapter on-screen LANGUAGE detection half (V2). BLOCKED ON CAPABILITY, on nobody's decision.** Tesseract OSD is the only detector this toolchain offers and it was measured wrong: three synthetic fixtures whose ground truth is Latin returned mixed-case -> Latin (confidence 5.15), source code -> Latin (7.41), and **ALL-CAPS -> CYRILLIC (18.33)**. The wrong answer arrived with the HIGHEST confidence of the three, so a confidence floor selects for the error instead of filtering it — and all-caps is not exotic in a screen recording, it is headings, UI labels and terminal banners. A fourth measurement bounds it further: OSD exits 1 on a sparse frame ("Too few characters"), and a frame with little text is the NORMAL case when sampling a recording — that outcome is could-not-determine and is never "nothing was on screen". What would unblock this half is a per-language recognition bake-off scored against **T132**'s hand-truthed sample, so it is downstream of T132 rather than of any decision. The probe carries the limitation as an `advisory` row that reproduces the measurement on every run and can never move the exit code

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T124 [UNBUILT] [REVIEW] Settle **U6** — what sampling period keeps an on-screen visibility — **BLOCKER: operator decision, purely** — every input is already recorded (segment median 6.74 s, p95 10.78 s, max 20.22 s) and frame sampling is measured working. What is missing is the operator CHOOSING a sampling period, which the task forbids rounding to a convenient number · **OWNER: operator** — unblocks T126/T127/T128 interval **no coarser than the segment precision it sits beside**? Measure it against the recorded segment distribution (median **6.74 s**, p95 **10.78 s**, max **20.22 s**); do not pick a round number. Three-valued exit. Record under `workshop/evidence/`. **T126 and T128 are bounded by this** — an interval bound chosen before it is measured is the same class of guess the no-threshold rule forbids elsewhere (FR-021, FR-055, FR-061) **BLOCKED ON OPERATOR — measured 2026-09-03.** U6 is a clarification and settling it is the operator's decision, not an agent's. Nothing technical is waiting: the inputs are already recorded (segment distribution median 6.74 s, p95 10.78 s, max 20.22 s) and the sampler is no longer hypothetical — `pipeline/detect_ocr.sh` has now measured frame sampling working on this host. What is waiting is the choice of period, which this line explicitly forbids rounding to a convenient number

▣

T125 [TDD] [REVIEW] Gate **G-OCR-2** — declare `screen_text` as a **passage kind** in the registry contract per `data-model.md` §2.9: minted through the **same** minter, resolved through the **same** four-outcome resolver, ordering key = visibility onset, provenance `ocr` and distinct from `asr`, engine confidence carried, redaction flag inherited. **Paired mutation**: mint an OCR passage through a second minter or a second identifier format; the gate must go red. **A fifth kind, not a fifth registry** (FR-060) (FR-007, FR-023, FR-060) **BUILT 2026-09-03.** `passagestore.KindScreenText` in `workshop/platform/backend/internal/passagestore/domain.go`, with gate **G-OCR-2** and its paired mutation in `screentext_test.go`, registered in `platform/gates/check-registry-002.tsv` as `G-OCR-2` and `T125-screen-text-paired-mutation` (registry rc **0**, 59 checks, 0 missing) (source). **A PRIOR NOTE ON THIS LINE WAS WRONG AND IS WITHDRAWN, NOT SILENTLY REPLACED.** It said "the registry it must extend is `platform/backend/pkg/search/`". That is the SEARCH INDEX's kind list, not the passage registry. §2.9's table is the field-by-field shape of `passage.Record`, so the registry is the passage registry and its vocabulary lives in `internal/passagestore/domain.go` — the one place the other four kinds are declared. Building it in `pkg/search` would have declared a corpus kind in the retrieval layer, which is the fifth-registry mistake wearing different clothes. The concurrent-editing reason given for deferring was therefore also moot: the file that needed editing was never contended. **What was added.** Five invariants beyond the existing F4/F5, each enforced on the library's own write path through `passage.WithRecordValidator` — so they run at `Put`, at `Sync` and at `Load`, not in a wrapper somebody can forget. **F5 was EXTENDED rather than duplicated**: `screen_text` joins `transcript_segment` as media-backed via `IsMediaBacked`, and the two differ in exactly one respect, deliberately — a transcript segment needs start **strictly before** end because speech has duration, while a `screen_text` permits onset == offset, because text seen at ONE sample of the grid has coincident observed bounds and forcing a wider interval would make the record assert a duration nothing measured. **F6** producer `ocr`, enforced in BOTH directions. **F7** interval bound present, finite, strictly positive — FR-061's "never zero" is refused by name, because a zero bound is not a tight measurement but a measurement never taken presented as the tightest possible one. **F8** engine confidence carried, `[0,1]`, caller-normalised so one scale reaches the registry. **F9** no speaker on on-screen text. **F10** an empty recognition is not a passage, guarded by `!Redacted` so it cannot block `RedactionLog.Purge` — the same tension the library resolves between its own F2 and F6, resolved the same way. **`producer` is NOT `passage.Provenance`, and conflating them**

was the near-miss. The library's field records whether the text is still the machine's or has been human-corrected. A corrected OCR passage is `human_corrected` **and** `ocr`; one field cannot carry both, and collapsing them would have reclassified every correction as a change of engine. **Proof, verified in BOTH directions — which a green run alone cannot show.** Green: **13 mutations, 13 caught, 0 missed** (in_process), with the **M0 negative control passing FIRST** (one of this tree's two shipped inoperative proofs failed precisely by having a broken control, so zero mutations ever ran). Red against two deliberately weakened throwaway copies: **W1** gutted `validateScreenText` — 7 uncaught, 1 caught by an unrelated rule (in_process), the load-at-rest mutation uncaught; **W2** changed one comparison so a zero bound was permitted, and **exactly M8** went red. W2 is the one that matters: a proof detecting only total removal would not have discriminated it. `domain.go` was restored to its pre-weakening sha256 and re-measured green. **The detect_ocr.sh acceptance rule was copied deliberately.** That probe compares output to KNOWN FIXTURE TEXT and refuses "non-empty" as a criterion, because a recogniser aimed at the wrong script returns confident nonsense rather than silence. Here every positive assertion compares a known VALUE, and every mutation must be refused with a message **naming the invariant it broke** — W1 proved that arm live by reporting `CAUGHT BY THE WRONG RULE` on M12 rather than counting a refusal it had not earned. **A hole this change opened, and closed.** Gate **G-SUG-6** (catalog and passage corpus are disjoint, so redaction has ONE authority) asserted disjointness from a HARDCODED list of four kinds. A fifth kind would have left a catalog row claiming `screen_text` passing. The list is now DERIVED from the new `passagestore.PassageKinds`, so the next kind is covered the day it is declared. Re-measured: G-SUG-6 green, 9 rows swept (in_process). **`screen_text` is deliberately NOT added to `baseIndexedKinds`.** Nothing produces one yet. `service.go`'s own comment records `diagram` as an existing advertised-but-unretrievable kind and forbids adding a second instance of that defect; the `T057-lesson-section-unindexed` registry row is the same precedent. Advertising a kind no corpus holds is a capability claim with nothing behind it. **Honest boundary.** No `screen_text` passage exists in the corpus — this task declares the KIND, and T126 (blocked on T124, an operator decision) is what produces one. All fixtures are synthetic; nothing was quoted from any recording. **Pre-existing failures, measured rather than assumed (in_process).** Four backend tests failed when this work began — `TestGateSUG4_RedactedPassagesNeverSurface`, `TestGateKG11_AdvertisedAreaTermKindsAreRetrievable`, `TestC4_1_5_AreaFilterEchoedAndApplied`, `TestC4_2_1_SuggestOffersAreaAndTer `mNames`. Each was measured **identical with and without this change** (HEAD’s `domain.go` restored, this task’s test file held aside), so none is caused by T125. **The “four” figure is now stale and is corrected rather than left standing: three were fixed by another agent’s concurrent work during this pass, and ONE remains — `TestGateSUG4` in `internal/api`.** It is neither caused nor fixed here. **`platform/gates/verify-check-registry-001.sh` exits 1 (source) on this tree, and it is NOT this task’s.** All 11 R5 violations are unregistered `*.py`/`*.sh` files under `pipeline/the_platform/`, landed by concurrent work; that registry’s `pipeline` scanroot sweeps recursively and is catching them exactly as designed. This task added **no** `*.sh` or `*.py` anywhere under `pipeline/`, so it has zero R5 exposure. They are deliberately NOT registered here: registering a gate whose paired proof has not been verified would be the bluff that registry’s own header forbids. **T126–T141 no longer sit behind this**

- [ ] T126 [UNBUILT] [TDD] Implement sampling and recognition across a chapter recording, producing — **BLOCKER: predecessor T124** — sampling cannot be implemented before the sampling period is chosen · **OWNER: implementer**, after T124 text plus a **visibility interval** and its **interval bound** (FR-061). The sampling period is a **recorded parameter** the run writes into its evidence, never a literal in the code — and the bound is never omitted and never presented as zero. Sources open read-only per **S7** (FR-061) **BLOCKED ON T124 (operator) ALONE — re-measured 2026-09-03, narrowed from “T124 AND T125”.** The toolchain half is no longer in doubt: `pipeline/detect_ocr.sh` exits 0 and proved frame sampling plus per-word recognition with confidence and geometry on this host, so nothing here is blocked on a capability. **T125 is now BUILT**, so the passage kind, its `ScreenTextObservation` constructor and its `interval_bound_s` field are all waiting for this task rather than the other way round. What is still missing is the sampling period, which T124 must settle and which this task is forbidden to hardcode — note `ScreenTextObservation` takes it as a PARAMETER precisely so no literal can be smuggled in here

- [ ] T127 [UNBUILT] [TDD] Gate **G-OCR-3** — implement the **stability rule**: text visible across — **BLOCKER: predecessor T124 + T126 · OWNER: implementer**, after T124/T126 consecutive samples is **one** passage with one interval, not one per sample. **Paired mutation**: emit one passage per sample; the gate must go red. This is not tidiness — a static slide left up for two minutes would otherwise inflate an area's evidence by the sampling rate, and the inflated count is what publication decisions rest on (FR-060, FR-061) **BLOCKED ON T126 ALONE — re-measured 2026-09-03, narrowed from "T125 AND T126".** The stability rule operates over consecutive samples; **the passage kind now exists** (T125), so what is missing is a sampler that emits any. Not blocked on capability: the engine's `tsv` output carries the per-word geometry this rule needs to decide that two samples show the same text

- [ ] T128 [UNBUILT] [TDD] Gate **G-OCR-4** — join `screen_text` mentions to the **existing** — BLOCKER:** **predecessor T124 + T126** · **OWNER: implementer**, after T124/T126 two-valued** precision model (FR-062, contract §3 N1/N2): declare `segment`, carry the **interval bound** beside it exactly as `word` precision carries timing confidence (N3). **Paired mutations**: (a) introduce a third `precision` value; (b) declare `word` precision on an OCR mention. Both must go red — (a) because every consumer switches on two values, (b) because no per-word timing record produced that time (FR-061, FR-062) **BLOCKED ON T124 (operator) — re-measured 2026-09-03, narrowed from "BLOCKED ON T125, BOUNDED BY T124".** The kind exists now, and it already carries the field this task must propagate: `passagestore.AttrIntervalBoundS`, with `IntervalBound()` to read it and F7 guaranteeing it is present, finite and non-zero on every `screen_text`. What is still unsettled is the VALUE, which T124 owns. Not blocked on capability. **Note what T125 did NOT do here**: it put the bound on the PASSAGE. FR-061 also requires it on every MENTION derived from one, and that half is this task's

- [ ] T129 [UNBUILT] [TDD] Gate **G-OCR-5** — record **modality** (`spoken` / `on_screen`) on every — **BLOCKER: predecessor T126** — modality cannot be recorded on mentions that do not exist yet · **OWNER: implementer**, after T126 mention, derived from the passage kind and never guessed, and mark a subject evidenced **only** on screen as `on-screen-only`, the modality analogue of `uncertain-only` (FR-012, FR-063). **Paired mutation**: default the modality to `spoken` where it is unset; the gate must go red. A term the workshop displayed but never discussed is a different fact from one it taught, and a reader who cannot tell them apart will over-read the taxonomy (FR-012, FR-063) **NO LONGER BLOCKED ON T125 — re-measured 2026-09-03. Its one recorded blocker is gone: the passage kind exists. It was NOT built in the same pass, and the reason is measured rather than a preference.** Modality is "derived from the passage kind, never guessed", and the derivation needs the

KIND at the point a mention is minted. It is not available there: `MentionDraft` carries `Passage passage.PID` and nothing else about the passage, and `MentionDraft.Mint(m *passage.Minter)` takes no registry — so deriving the kind means either threading a `*passage.Registry` through `Mint` and every call site, or having `pipeline/mentions/join.py` emit modality in its JSONL. Both are real changes with real blast radius, and "never guessed" forbids the cheap third option of defaulting when the kind cannot be resolved — which is precisely what this task's own paired mutation targets. **It is also COUPLED TO T128, which is operator-blocked.** `Mention.Time` is documented as "nil exactly when Passage is not a transcript segment". A `screen_text` is now media-backed, so a mention on one SHOULD carry a time — its visibility interval, at `segment` precision, with the interval bound beside it. That is T128's shape, and T128 waits on T124 for the bound's value. Building modality first would freeze a mention shape that T128 must then change. **And its second half cannot be exercised against anything real.** Marking a subject `on-screen-only` needs a subject evidenced only on screen; there are ZERO on-screen mentions in the corpus and no path to one until T126 runs. A gate for it today could only ever run on synthetic fixtures, with its production path unreachable — which is the shape of a proof that passes without proving. **Reclassified: merely unwritten, but correctly sequenced AFTER T128, not before it.** Not blocked on capability

- [ ] T130 [UNBUILT] [REVIEW] Settle **U7** — the **corroboration window**. A term is commonly — **BLOCKER: operator decision** — the corroboration window (U7) is an unsettled parameter, same class as T124 · **OWNER: operator** displayed *before* it is discussed, so measure the observed lead/lag distribution between on-screen and spoken occurrences rather than assuming coincidence. Three-valued exit. Record under `workshop/evidence/`. **T131 is bounded by this**: a window tuned to zero lag under-groups, and a window widened until the numbers look tidy over-groups — and over-grouping destroys evidence silently (FR-055, FR-063) **BLOCKED ON CAPABILITY FIRST, THEN OPERATOR — measured 2026-09-03.** Unlike T124, this clarification cannot be settled from anything already recorded: it asks for the observed lead/lag distribution between on-screen and spoken occurrences, and **no OCR output exists in this tree to measure a distribution over**. T126 must run first. Settling U7 on the resulting distribution is then the operator's decision

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T131 [UNBUILT] [TDD] Gate **G-OCR-6** — implement **corroboration grouping** per — **BLOCKER: predecessor T130 · OWNER: implementer**, after T130 `data-model.md` §2.10: mentions of one subject whose times overlap within the measured window form one group, and every publication, coverage and attachment figure counts **groups**, never raw mentions (FR-063). Both figures are published together; **neither mention is deleted** and both stay individually retrievable and navigable. **Paired mutations**: (a) count raw mentions; (b) widen the window to the whole chapter. Both must go red — (b) because two genuinely distinct occurrences twenty minutes apart would collapse, which is how deduplication turns a corpus into a vocabulary list (FR-063, SC-035) **BLOCKED ON T130 — measured 2026-09-03.** The window is the parameter this task groups by, and it is unmeasured. Not blocked on capability

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T132 [UNBUILT] [TDD] Build the **hand-truthed ground-truth sample** — drawn by a recorded seed — **BLOCKER: irreducible HUMAN LABOUR over a private recording** — hand-truthing a ground-truth sample cannot be delegated to a machine, and this is the SINGLE HIGHEST-LEVERAGE operator item in spec 002: T133, T134, T135, T136, T142 and 002's own closure all sit behind it · **OWNER: operator / human labour** — nothing an agent can do so it is reproducible, with the **population size published** beside it, matching the sampling discipline T081 already uses. This artifact is the input to **both** accuracy axes; a figure published without its seed and population is not a measurement anyone can re-derive (FR-064, SC-031, SC-032) **BLOCKED ON OPERATOR — measured 2026-09-03, and this one cannot be automated away.** Hand-truthing is human labour, performed over a private recording of a teaching session with an identifiable third party. No agent can substitute for it here, and the obvious shortcut is the precise mutation **T133** requires to go red: a machine-produced "ground truth" scored against machine output measures nothing. The seed and population-size discipline this line asks for is already demonstrated by T081 and needs no new capability. **T133, T134 and — through them — the closure of spec 002 all sit behind this operator task**

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T133 [UNBUILT] [TDD] Gate **G-OCR-7** — measure **textual** accuracy per chapter: word error — **BLOCKER: predecessor T132** — textual accuracy is measured AGAINST the hand-truthed sample · **OWNER: implementer**, after T132 rate and character error rate against T132's sample, scored by the **same** edit-distance method the speech calibration already uses, so the two figures are comparable rather than merely adjacent (FR-064, SC-031). **Paired mutation**: score against the recogniser's own output instead of the ground truth; the gate must go red (FR-064, SC-031) **BLOCKED ON T132 (operator) — measured 2026-09-03.** There is no ground truth to score against. The edit-distance method this must share with the speech calibration already exists (`pipeline/compare_engines.py`), so the scorer is not the gap; the reference is

- [ ] T134 [UNBUILT] [TDD] Gate **G-OCR-8** — measure **temporal** accuracy per chapter: the — **BLOCKER: predecessor T132** — temporal accuracy is measured against the same sample · **OWNER: implementer**, after T132 proportion of on-screen mentions whose declared visibility interval **contains** the moment the text was actually on screen (FR-064, SC-032). **Paired mutation**: assert only that an interval exists; the gate must go red. **This axis is separate from T133 deliberately**: deep linking depends on this figure and on no other, and a perfect textual score is entirely compatible with every interval being wrong — a text-only measurement cannot see the failure at all (FR-064, SC-032) **BLOCKED ON T132 (operator) — measured 2026-09-03.** Same missing reference. Note the axis separation this line already argues for is now measurable in principle: `pipeline/detect_ocr.sh` confirmed the engine returns per-word bounding boxes, which is what a visibility interval is ultimately derived from

- [ ] T135 [UNBUILT] [TDD] Gate **G-OCR-9** — implement the **accuracy budget, derived and never** — BLOCKER:** **predecessor T132, AND a second independent operator blocker** — `CALIBRATION.md` carries NO WER figure at all, so the floor this gate must read at run time does not exist. The fix is spec 001's T037(b) blind human reference (30 seeded windows, 15 min audio, costed 1-2 h) — **one piece of human work satisfies both specs · OWNER: operator / human labour**, then **implementer** picked** (FR-064a): the floor is **read at run time** from the recorded speech calibration for the same corpus, and the gate publishes the OCR figures and the speech baseline **together** (SC-033). Three-valued — calibration record or ground-truth sample unreadable ⇒ **2**, never a pass. **Paired mutations**: (a) hardcode the floor as a literal; (b) measure accuracy over the frames the engine was tuned on. Both must go red. **This is how an accuracy budget is stated without violating the no-guessed-threshold rule**: U2 and U3 forbid guessing a target for a quantity nobody has measured, and this floor is not guessed — it is read from a measurement this project already has (FR-054, FR-064a, SC-033) **BLOCKED ON A MISSING MEASUREMENT — measured 2026-09-03 — and this is NOT merely "blocked on T133 and T134". The floor this gate is required to READ AT RUN TIME does not exist.** This task derives the budget from "the recorded speech calibration for the same corpus", and that record was checked rather than assumed: `workshop/pipeline/CALIBRATION.md` carries **no WER figure**. Its own U4 row reads *"achievable WER | open — needs §5's blind human reference"*, and it states expressly that the engine-to-engine divergence (3.19 %) **is not a WER and must never be quoted as one**. So even after T133 and T134 produce OCR figures, there is nothing to compare them against, and the one number lying nearby is explicitly disqualified — quoting it would be precisely the substitution this line's mutation (a) forbids. **Unblocking needs the blind human reference of CALIBRATION.md §5, which is operator labour**, in addition to T132/T133/T134. Recorded here because the task as written reads as though its input were already in hand, and it is not **THE BLOCKING INPUT IS NOW SPECIFIED, SEEDED AND HANDED OVER — under spec 001's T037, not here. Updated 2026-09-03; still BLOCKED, and the block is unchanged operator labour, but it is no longer unattributed.** The blind human reference CALIBRATION.md §5 needs is exactly the reference `specs/001-workshop-curriculum-platform/tasks.md` **T037(b)** specifies: 30 seeded 30 s windows over chapter 01, 900.0 s = 15 min of audio, listed with exact `t0`/`t1` bounds in `workshop/chapters/01/transcript/accuracy-plan.json`, costed at **1–2 hours**. **ONE piece of human work satisfies both 001's SC-002 and this task's FR-064a floor** — it is not two transcription jobs, and it must not be scheduled as two. The plan was re-emitted on 2026-09-03 under `research/transcription.md` §5.2's **ten equal temporal strata** (3 windows each, occupancy `[3,3,3,3,3,3,3,3,3,3]`, 0 empty, 0 overlapping), superseding a confidence-stratified set; a reference transcribed against the superseded windows would match nothing. Running `bash workshop/scripts/verify-accuracy.sh 01 --reference <path>` writes `workshop/chapters/01/transcript/accuracy.json`, and **that file is the record this gate reads at run time** — seeded, reproducible, population published, scored by the **same** `pipeline/compare_engines.py` edit distance T133 must share, which is what makes SC-031's *"comparable rather than merely adjacent"* structural instead of coincidental. **A SAMPLED estimate satisfies FR-064a — checked, not assumed.** FR-064a says *"read from the recorded speech-recognition calibration"* and never says exhaustive; FR-064 makes the OCR side itself a **sample**; and this task's own three-valued clause says *“ground-truth **sample** unreadable ⇒ 2”*. **A whole-chapter WER is NOT required.** Had it been, T037 would not close this and that would be recorded here instead. **THREE THINGS THIS DOES NOT GIVE YOU, named so they are not discovered late.** (1) **Scope:** FR-064b/SC-034 are per chapter and this reference covers **chapter 01 only** — no other chapter acquires a floor. (2) **CER:** SC-031 requires the OCR figure as WER **and** CER; `verify-accuracy.sh` computes **WER only**. Producing a speech CER needs **no additional human work** — the same reference text yields it — but it needs a **T112 change** in 001, and without it SC-033 publishes an asymmetric pair. (3) **Temporal symmetry:** T134/SC-032 measure a temporal axis; the speech-side analogue is 001’s timestamp-error companion metric (SC-003), whose only possible input is the `onset_s` mark the reviewer makes **while listening**. T037’s handover asks for it as an optional field for exactly this reason — skipped, it costs a second 1–2 hour listening pass. SC-033’s wording does not demand it, so its absence does not fail this gate; it forecloses a symmetry that is nearly free today. **THE ESTIMATOR UNIT, if any interval is ever put on either figure: the WINDOW / SAMPLE UNIT (n = 30 clusters), never the word.** Words inside one 30 s window are not independent draws, and WER is not a proportion — insertions let `(S+D+I)/N_ref` exceed 1, which no binomial admits. At the limit of perfect intra-window correlation a word-level interval is narrower by up to `√(N_hyp/30) ≈ 8×`, **overstating** precision. **Nothing in this project computes an interval today**, and `accuracy.json` carries no interval field — it carries this warning in `sample.estimator_unit` instead

- [ ] T136 [UNBUILT] [TDD] Gate **G-OCR-10** — implement the **per-chapter publication precondition** — **BLOCKER: predecessor T133 + T134 + T135** — a publication precondition cannot be enforced before the figures it gates on exist · **OWNER: implementer**, after the accuracy chain (FR-064b): no OCR-derived mention is published for a chapter until **that chapter’s** accuracy run has been observed passing. **Paired mutation**: publish from a chapter whose accuracy run returned could-not-determine; the gate must go red. Font, resolution and compression differ between recordings, so a figure measured on one chapter is not evidence about another — the same per-chapter discipline **V2** already applies to word timings (FR-064b, SC-034) **BLOCKED ON T135 — measured 2026-09-03.** There is no budget to precondition publication on. Not blocked on capability

- [ ]

T137 [UNBUILT] [TDD] Gate **G-OCR-11** — extend the content-boundary check over OCR output and — **BLOCKER: predecessor T126** — there is no OCR output to extend the content-boundary check over · **OWNER: implementer**, after T126 run it **before** publication (FR-065, SC-036). **Paired mutation**: scope the check to transcript text only; the gate must go red. **On-screen text is a wider disclosure surface than the transcript** — a recording displays window titles, file paths, identifiers and names that nobody ever said aloud, so a boundary check written against spoken text does not cover this kind. The umbrella repository is public and this material is not (FR-065, SC-036) **BLOCKED ON T125 AND T126 — measured 2026-09-03.** The check must run over OCR output and none is produced yet. Worth restating rather than deferring silently: this task guards the widest disclosure surface in the phase, and the umbrella repository is public while this material is not, so it must land BEFORE T141 wires the stage into the chapter-addition path — not after

- [ ] T138 [UNBUILT] [TDD] Index `screen_text` on its own text and prove **G-KG-11** for it — an — **BLOCKER: predecessor T126** — nothing to index until `screen_text` mentions are produced · **OWNER: implementer**, after T126 advertised kind must be **retrievable**, proven by a planted known target, **never** by a row count. Do not advertise the kind until it is retrievable (FR-066). The corpus already advertises `diagram` with **0** entries, and this task exists so a second such kind is not created (FR-028, FR-066, SC-013) **BLOCKED ON T125 — measured 2026-09-03.** Nothing to index until the kind exists. Not blocked on capability

- [ ] T139 [UNBUILT] [P] Add the contract sections and route-manifest rows for the `screen_text` — **BLOCKER:** none but the work — contract sections and route-manifest rows can be written ahead of the implementation, and doing so early is cheap · **OWNER: implementer** — unblocked today kind, its evidence entries and the modality and interval-bound fields (FR-059). Gate **G-KG-1**. This task is also what moves gates **G-OCR-1**..**G-OCR-11** into `contracts/`, where the gate-attachment closure check enumerates them — they are attached to task lines already, so that check must still print `unattached: 0` after this task lands (FR-059) **MERELY UNWRITTEN, AND DELIBERATELY NOT DONE IN THIS PASS — measured 2026-09-03.** No capability and no operator decision blocks the contract text. It was left alone anyway, because doing it first would move eleven `G-OCR-*` ids into `contracts/` while **none of the eleven**

gates exists, publishing gate identifiers for gates nothing builds. The gate-attachment closure check would still report `unattached: 0` — every id is already carried on its task line — so the check would stay green while the contract described a capability the tree does not have. That is a green instrument reporting on the wrong question, and the correct order is gates first, contract second. Measured before and after this pass with the corrected extractor: **19 ids, `unattached: 0`**, unchanged

- [ ] T140 [UNBUILT] [TDD] Prove redaction reaches **OCR-derived** mentions across all eight targets — **BLOCKER: predecessor T126** — redaction cannot be proven to reach OCR-derived mentions that do not exist · **OWNER: implementer**, after T126 in `data-model.md` §5. Gate **G-KG-7** extended. **Paired mutation**: restrict propagation to the spoken modality; the gate must go red. The eight targets do not change — what must be proven rather than assumed is that the propagation was not written against ASR-derived mentions only (FR-027, SC-012) **BLOCKED ON T125 AND T126 — measured 2026-09-03.** There are no OCR-derived mentions for redaction to reach. Not blocked on capability

- [ ] T141 [UNBUILT] [US7] Add the OCR stage to the platform's **existing** chapter-addition path per — **BLOCKER: predecessor T126 + T136** · **OWNER: implementer**, after T126/T136 **S1** — a stage added, **not a second procedure**; three-valued and resumable per **S3/S4**; writing to no source per **S7**; writing evidence on every outcome including could-not-determine per **S8**. Gate **G-KG-17** extended to assert the OCR outputs appear in the synthetic-chapter run. **Paired mutation**: remove the OCR stage; the gate must go red naming the missing output (FR-033d, FR-033e) **BLOCKED ON T126 — measured 2026-09-03.** There is no OCR stage to add to the chapter-addition path. Not blocked on capability: the two tools the stage would call, frame sampling and recognition, are both measured working by `pipeline/detect_ocr.sh`

- [ ] T142 [UNBUILT] [REVIEW] Publish both OCR accuracy figures **beside** the speech-recognition — **BLOCKER: predecessor T133 + T134, and spec 001's T037** — both OCR accuracy figures must be published BESIDE the speech-recognition figure, and that figure does not exist either · **OWNER: implementer**, after T133/T134 and 001's T037 ones in the shipped limits document and state plainly what OCR cannot do (FR-050 discipline); register every check added by this phase (SC-027); and drive each of **G-OCR-1**..**G-OCR-11** into its **could-not-determine** condition and assert the third state (SC-028). A figure published alone is not an accuracy claim a reader can act on, and a gate never observed in its third state is not known to have one (FR-050, SC-027, SC-028, SC-033) **BLOCKED ON EVERY TASK ABOVE — measured 2026-09-03.** This is a final review task and it cannot honestly close while any of T123-T141 is open. One part of it is already partly satisfied and should not be re-counted later: **G-OCR-1** has been driven into its could-not-determine condition and the third state asserted — `pipeline/detect_ocr.sh --scratch-dir /nonexistent/detect-ocr-probe` exits **2**, executed on every default run of `platform/gates/verify-check-registry-001.sh` as that row's registered rc-2 probe. The other ten `G-OCR-*` gates do not exist, so nine tenths of this obligation is untouched

- [x] T143 Build **G-KG-1-changed** and its §1.1 paired mutation: assert that every §4 *changed* endpoint carries a route-manifest row whose contract citation names **both** its `001/contracts/http-api.md` section and its §4 subsection here (R1b), and prove the gate red by **rewriting** one such citation back to its 001-only form — leaving the row otherwise byte-identical. The mutation must be a rewrite, not a deletion: `G-KG-1`'s deletion mutation cannot reach this defect, because every §4 row is **present** throughout and the failure is a row that is present but under-cites (FR-059). **PARTIAL, 2026-09-03 — the DEFECT is fixed and the box is unticked, but the stated reason has been CORRECTED the same day and the correction makes the residue smaller and sharper.** The compound form already existed on three of the four changed endpoints (`3.6+002.4.2`, `3.7+002.4.1`, `3.11+002.4.4`); §4.3's answering row carried a bare `3.10` and now reads `3.10+002.4.3`, so R1b is met 4 of 4 as measured that day (`verify-server-unity.sh` re-run after the edit: `PASS=35 FAIL=0 UNDET=0 DEBT=4`, exit 0 — population: unstated).

```
**WITHDRAWN, not restated: *"the GATE is not built"* and
*"Nothing enforces it."*** Both were
wrong when written and the measurement is one command:

```bash
grep -n 'G-KG-1-changed' workshop/platform/gates/check-
registry-002.tsv
#   29: check  G-KG-1-changed  go-test   # (source)
#       internal/api/
knowledge_gates_test.go::TestGateKG1_ChangedEndpointsCiteTheDel
cd workshop/platform/backend
go test -count=1 -run
TestGateKG1_ChangedEndpointsCiteTheDeltaContract ./internal/
```
```

api/... # ok (in_process)
```

A gate carrying this exact id exists, is registered, is green, and its §1.1 pair
(`platform/backend/gates/prove-knowledge-manifest-mutation.sh`) performs the **rewrite**
mutation this task specifies rather than a deletion — its own header says it *"rewrites one of
the three contract columns back to its bare specs/001 token in the REAL production manifest,
requires this test to go RED with its own message, restores the file byte-identically and
requires it GREEN again."* All of that landed in workshop `bd7da41`, before this task was
written.

**What is genuinely still owed — and it is the one endpoint this task was created for.** The
existing gate's `want` table enumerates **three** rows: `GET /api/search` (3.7+002.4.1), `GET
/api/suggest` (3.6+002.4.2) and `GET /api/progress` (3.11+002.4.4). Its own doc comment scopes
itself that way, naming §4.1, §4.2 and §4.4 and attributing itself to **T070**. **`/api/ask`
— §4.3, the fourth changed endpoint and the only one whose citation was actually wrong — is
absent from the table.** So the gate covers 3 of 4, and the row whose `3.10 → 3.10+002.4.3`
repair is recorded above is enforced by nothing: the next edit that drops that suffix still
restores the violation silently.

~~**This box therefore stays `[ ]`, and the remaining work is now a small, named change rather
than a gate from scratch:** add the `/api/ask` row to the existing table (and, per §1.1, a
fourth rewrite case to `prove-knowledge-manifest-mutation.sh`, whose current mutation set does
not touch it).~~ Do not tick this on the strength of the gate existing — a gate that enumerates
three of four obligations is green about the three and blind to the fourth, which is the same
blind-instrument shape the closure check's own `THE BLIND ZERO` note records one section
below. **That caveat is retained because it is the standing lesson, and it is now SATISFIED
rather than waived — the gate enumerates 4 of 4.**

**TICKED 2026-09-03. The residue named above is closed, and it was closed by DERIVING both
coverage sets rather than by extending either hand list — which
```

is a stronger fix than this
task asked for, and deliberately so.** A hand list is what produced the defect: the table
enumerated §4.1, §4.2 and §4.4 and omitted §4.3, and nothing anywhere was checking that
omission. Adding a fourth literal would have left the next §4.5 exposed to the identical
failure. Measured, both directions:

- **The gate.** `TestGateKG1_ChangedEndpointsCiteTheDeltaContract` no longer carries a `want`
  table. It locates `contracts/http-api-delta.md` by a parent walk mirroring
  `verify-server-unity.sh`'s own `find_contract()` (override:
  `WORKSHOP_HTTP_DELTA_CONTRACT`), parses every `### 4.N` heading under `## 4. Endpoints
  changed`, and grades each one. It PRINTS the derived population so a green run shows what it
  graded: *"coverage set DERIVED from …/http-api-delta.md: 4 changed endpoint(s) — §4.1 GET
  /api/search, §4.2 GET /api/suggest, §4.3 POST /api/ask, §4.4 GET|POST /api/progress"*.
  Matching is on PATH with the query string stripped, exactly as U6 does, because §4.3's
  heading says `POST /api/ask` while the probed row is `GET /api/ask?q=ping` and §4.4's says
  `GET|POST` — requiring the verbs to agree would fail on a disagreement that is not a defect.
  The delta citation is compared as a whole `+`-joined COMPONENT, not by substring, so
  `002.4.1` can never be satisfied by `002.4.10`.
- **The proof.** `prove-knowledge-manifest-mutation.sh`'s `CHANGED_*` arrays are gone too; its
  §4 mutation set is derived from the production manifest (every row whose contract column
  carries a `002.4.N` component). `--list` now reports **12 mutations (8 fixed §3 deletions +
  4 DERIVED §4 citation rewrites)**, the fourth being `ask-4-3: rewrite /api/ask?q=ping
  contract column 3.10+002.4.3 -> 3.10`. Full run: **rc 0, `12 proven / 0 problem / 0
  undetermined`** (in_process) — so the `/api/ask` citation is now enforced by a mutation that has been
  OBSERVED turning the gate red, which is what the repair recorded above previously lacked.
- **The two derivations are taken from DIFFERENT documents on purpose.** The gate derives from
  the contract's §4 headings; the proof derives from the manifest's own citations. A row that
  loses its citation drops out of the proof's set but turns the gate RED, and the proof's

```
  preflight refuses to prove anything against an already-red
gate (exit 2). Agreement is the
  evidence; disagreement is an UNDETERMINED, never a silent
under-mutation.

**Honest boundary (§11.4.6), stated because it is a real change
in behaviour.** The gate now
reads a document that lives in the SUPERPROJECT. Where it
cannot be found, the test reports
**UNDETERMINED via `t.Fatal`, i.e. as a FAILURE** — `t.Skip`
was rejected because a skip exits
0 and `check-registry-002.tsv` would read it as a pass. So a
workshop checkout taken apart from
its superproject now fails this one test instead of passing it
vacuously. That mirrors
`verify-server-unity.sh` U6, whose own comment says an
unreachable contract "is exit 2, not a
pass"; the previous test comment argued the opposite and is
withdrawn in the file itself.

**The superseded "measured while re-deriving" note below is
retained, not deleted** — it
recorded that the paired proof could not be re-run to
completion because of a concurrent Go
build break. It has since been re-run to completion (rc 0,
above). The note stands as the
record of a COULD-NOT-DETERMINE that was correctly refused as a
pass.

**Measured while re-deriving this, and recorded rather than
absorbed: the paired proof could
NOT be re-run to completion today** — `bash platform/backend/
gates/prove-knowledge-manifest-mutation.sh`
returned rc=1 with `7 proven / 1 problem / 1 undetermined`
(in_process), and the problem and the
undetermined are both a **Go build break from concurrent
editing in this tree**
(`internal/redaction/taxonomy.go: t.ProposalsUnlinked
undefined`, and earlier
`pkg/index/generation.go:170: declared and not used:
declined`), not a defect the proof found.
That is a COULD NOT DETERMINE on the proof's own health, and a
2 is never a pass; the gate
itself was green minutes earlier on the same tree
```

Checkpoint: on-screen text is addressable, counted once where it coincides with speech, and its accuracy is **measured and published beside the speech baseline**. **This is the checkpoint that closes spec 002** — until T133, T134 and T135 have produced figures, this feature is not finished regardless of how many checkboxes above it are ticked.

Dependencies

- **Phase 2 blocks everything.** Every later artifact points at something through the identity, precision and resolution mechanism built there.
- **Phase 3 blocks Phases 4–9.** There is nothing to link, search, assess or export until areas exist.
- **Phase 4 blocks Phases 5–7.** Search locus, question jumps and the surface all navigate through the graph.
- **Phase 6 blocks Phase 7's practice view** (T087).
- **T059 blocks T060.** Offsets must not be implemented on an assumption about the index.
- **T114 blocks T115 and T116. Discharged 2026-09-02** — T114 ran and settled U4 (the entailment judge produced a DECIDED verdict on this host). ~~T115 and T116 are unbuilt, not blocked.~~ **Superseded 2026-09-03: both are BUILT and ticked.** The whole chain — U4 measured, decision taken, verifier built, gates registered and green, live on the deployment — is closed.
- **T085 blocks T091.** The before-list must exist before the reshaping it measures.
- **T004 blocks T097 and T098.**
- **Phase 11 is sequenced after every unbuilt task above it.** On-screen mentions attach to areas, feed coverage figures and export alongside spoken ones, so a Phase-3-to-Phase-10 defect reached through OCR-derived evidence would be diagnosed twice.
- **T124 bounds T126 and T128** (the sampling period, hence the interval bound). **T130 bounds T131** (the corroboration window). Both are `[REVIEW]` measurements with three-valued exits, and both exist so a number is measured rather than picked.
- **T125 blocks T126–T141.** Nothing may produce, join, count or index a `screen_text` passage before the kind exists in the registry contract.
- **T132 blocks T133 and T134**, which together block **T135**, which blocks **T136**. The budget cannot be evaluated before both axes are measured, and nothing may be published before the budget is evaluated.
- **T133, T134 and T135 block the closure of spec 002 itself.** This is the dependency D5 added.
- **T031 is blocked on the prose-authorship clarification** — and on nothing else, and it is now the ONLY `[BLOCKED: …]` marker left in this file; ~~**T094** on the UX research~~; ~~**T115/T116** on the answering clarification~~ — **both withdrawn 2026-09-02**: the answering clarification is decided and U4 is measured, so T115/T116 became `[UNBUILT]`, not `[BLOCKED]` — **and on 2026-09-03 that marker was**

discharged too, because the code was written; T094's blocker was resolved by a real ranked, sourced research artifact (`workshop/docs/research/ux-research-2026-09-01.md`), and T060's was discharged when T059 settled U1 (`workshop/docs/session-evidence/phase5-report.md`).
- **The rule this section used to state — *"Markers record what a task waited on, not what it is still waiting on — read the checkbox for that"* — is WITHDRAWN, not restated.** It was a rule written to excuse two stale markers rather than to remove them, and it made `[BLOCKED: …]` unreadable: under it, a marker asserted nothing a reader could act on, and the only way to learn whether a task was actually held up was to ignore the marker. **The markers on T060 and T094 were stripped on 2026-09-02 instead**, with the discharge and its evidence recorded in each task's own note, so the history survives without the standing claim. The rule in force now is the plain one: **a `[BLOCKED: …]` marker means the task is blocked TODAY.** Discharge it by removing the marker and recording what discharged it — never by redefining what the marker means.

Parallel opportunities

- Taxonomy extraction (Phase 3) and the front-end surface (Phase 7) — different languages, different directories, joined only by the knowledge contract.
- Question authoring (T075) and the export pipeline (Phase 8) — export consumes documents, not the process that wrote them.
- Documentation (T121) and the behavioural gates, once the contracts are fixed.

Independent test criteria

Story	Deliverable that stands alone
US1	the taxonomy and the area documents — readable and citable with no interface
US2	every relationship traversed in reverse through the platform's own interfaces
US3	the benchmark run, with per-query outcomes and locus assertions

Story	Deliverable that stands alone
US4	every served question enumerated and every citation resolved
US5	each view opened and completed by keyboard, audited, token-checked
US6	the format matrix and the re-export text diff
US7	the synthetic-chapter run and its diff
Phase 11	the two OCR accuracy figures, published beside the speech-recognition ones

Gate coverage — every contracted gate has a task that builds it

Gate	Task
G-KG-1	T040, T054, T070, T083
G-KG-2	T072
G-KG-3	T073
G-KG-4	T012
G-KG-5	T060
G-KG-6	T019
G-KG-7	T021
G-KG-8	T016
G-KG-9	T098
G-KG-10	T024
G-KG-11	T058
G-KG-12	T079
G-KG-13	T008
G-KG-14	T028
G-KG-15	T012
G-KG-16	T076
G-KG-17	T109, T141

Gate	Task
G-KG-18	T110
G-KG-1-changed	T143

19 gates, 19 paired mutations, all owed. The `18 gates, 18 paired mutations` figure this line carried is WITHDRAWN, not restated — it was the population the OLD, blind extractor could see, and the nineteenth id (`G-KG-1-changed`, defined in `contracts/http-api-delta.md` §5) was missing from this table for the same reason it was missing from the closure check: an extractor that stopped at the digits read it as `G-KG-1`. See `THE BLIND ZERO` below. A gate never observed failing is not known to work, and every proof must include a case that runs the real entry point against the real tree.

On-screen text gates — added 2026-09-02, and NOT yet in `contracts/`

Gate	Task	Subject
G-OCR-1	T123	OCR toolchain capability-probed, not assumed from a name on `PATH`
G-OCR-2	T125	`screen_text` mints through the one existing minter
G-OCR-3	T127	one passage per contiguous visibility, never one per sample
G-OCR-4	T128	precision stays two-valued; interval bound carried beside it
G-OCR-5	T129	modality declared on every mention; `on-screen-only` marked
G-OCR-6	T131	evidence counted over corroboration groups, never raw mentions
G-OCR-7	T133	

Gate	Task	Subject
		textual accuracy measured per chapter against a seeded truth sample
G-OCR-8	T134	temporal accuracy measured per chapter — intervals actually contain the text
G-OCR-9	T135	budget read from the speech calibration, never a literal; three-valued
G-OCR-10	T136	nothing published from a chapter whose accuracy was not measured
G-OCR-11	T137	OCR output passes the content boundary before it reaches a public artifact

Eleven gates, eleven paired mutations, all owed and none built.

Honest boundary on the closure check — read this before quoting its green result. The gate-attachment closure check enumerates gate ids from `contracts/`, and `contracts/` is **not** modified by this change. It still defines exactly the **18** `G-KG-*` ids, so the check's population is unchanged and it reports **`unattached: 0`** — measured 2026-09-02 both before and after this phase was written. **That zero is a true result and it is not evidence about these eleven gates: they are not in its population yet.** A green check here says nothing whatever about `G-OCR-*`, and reading it as though it did would be exactly the blind-instrument error the standing rules name.

T139 is the task that puts them into the population, and the forward guarantee was made checkable rather than promised: all eleven ids were placed on the **task line itself** — not on a continuation line — and that was verified, so the check must still report `unattached: 0` the moment T139 lands, under **either** reading of "attached". Measured

2026-09-02: 11 of 11 `G-OCR-*` ids satisfy the strict line-anchored form. If a later run reports one unattached, the failure is real and belongs to whoever moved the id off its task line.

```
# THE CLOSURE CHECK, in the wrap-aware form this file's line-wrapping requires.
# A contract gate must be carried by a TASK BLOCK — the "- [ ] T###" line plus its
# indented continuation lines. Prints one row per unattached gate; silence = closed.
cd specs/002-knowledge-areas-deep-linking
awk '
  /^- \[[ xX]\] T[0-9][0-9][0-9]/ { if (blk!="") print blk; blk=$0; next }
  /^## /                          { if (blk!="") print blk; blk=""; next }
  /^[[:space:]]*$/                { next }
  /^[[:space:]]/                  { if (blk!="") blk = blk " " $0; next }
  { if (blk!="") { print blk; blk="" } }
  END { if (blk!="") print blk }
' tasks.md > /tmp/blocks.txt
miss=0
for g in $(grep -rhoE 'G-[A-Z]+-[0-9]+(-[a-z]+)*' contracts/ | sort -u); do
  grep -qE "${g}([^0-9A-Za-z-]|$)" /tmp/blocks.txt || { echo "UNATTACHED $g"; miss=$((miss+1)); }
done
echo "unattached: $miss"
        # 19 ids. On 2026-09-03 the corrected extractor
                          # first reported ONE — G-KG-1-changed; it is now
                          # attached to T143, so this reads 0. The ONE is
                          # what the old extractor could not see at all.
```

> **THE BLIND ZERO, found 2026-09-03 — and this file is where it actually bit.** Feature 001's companion section records the general lesson; this is the instance. Until that date both regexes above were `G-[A-Z]+-[0-9]+` and `([^0-9]|$)`, and this check printed `unattached: 0`. **That zero was blind, not clean.** The extractor stops at the digits, so the id `G-KG-1-changed` — defined in `contracts/http-api-delta.md` §5 — was read as **`G-KG-1`**, a different gate that *is* attached, to T040. The compound id never entered the loop, so it could be neither attached nor reported. The corrected extractor sees **19** ids where the old one saw 18, and the nineteenth reports `UNATTACHED G-KG-1-changed`.

> **What the blind zero was hiding is worse than an unattached id.** `G-KG-1-changed` asserts R1b: every §4 *changed* endpoint carries a manifest row citing **both** its 001 section and its §4 subsection here. Measured the same day, the compound citation form already exists and three of the four changed endpoints carry it — `3.6+002.4.2` (`/api/suggest`), `3.7+002.4.1` (`/api/search`), `3.11+002.4.4` (`/api/progress`). **§4.3's rows carried a bare `3.10`.** So the gate nobody could see was a gate that would have been **red**, and the thing it would have caught was a real, standing contract violation — not a bookkeeping slip.
>
> T040 was **not** the right place to attach it, and attaching it there would have been the papering-over this section exists to forbid: T040 builds the manifest rows for the endpoints **new in §3**, whereas R1b governs rows that already existed from feature 001 and were only *amended*. The gate needs a task that owns that work — see **T143**.

> **Why the wrap-aware form, and why presence-counting is not the check.** Feature 001 records the durable lesson: a set-difference over identifiers appearing *anywhere* in the file went green the moment a paragraph **describing** the missing gate mentioned its id — documenting the defect fixed the check while the defect stood untouched. The check must therefore be anchored to a task, not to the file. It must also be anchored to a task **block** rather than a task **line**, because this file wraps its tasks across several lines: run against `tasks.md` in the single-line form 001 uses, five ids (`G-KG-1`, `G-KG-5`, `G-KG-7`, `G-KG-10`, `G-KG-12`) report unattached purely because their id fell past a line break. That is a measurement of the line-wrapping, not of the gates. The block form above is the same closure condition read correctly, and **both counts were re-measured after this phase was added and neither moved**.

Implementation strategy

Build Phase 2 completely and stop. It is the only phase whose defects are invisible — a wrong identifier scheme, a conflated precision or a collapsed resolution outcome all keep rendering links that look correct. Everything after it fails loudly by comparison.

Then take one area end to end — promoted, evidenced, authored, linked, searchable, assessed, exported — before applying the pattern to the rest. A pattern proven on one area is cheap to correct; the same mistake made across every area is a migration.

Phase 11 comes last, and its accuracy tasks come before its publication task. The ordering inside the phase carries the same logic as Phase 2's: measure the property that is invisible when wrong before anything depends on it. OCR that is 90% right looks identical to OCR that is 99% right in every interface built on top of it — the difference is only ever visible in a measurement, and only if that measurement was taken before the mentions were published rather than after somebody complained.

Task count: 143, contiguous `T001–T143`, verified by parsing this file rather than by counting. Of those, **104 are ticked and 39 are not** — **18** carried forward from `T001–T122` and **21 in the Phase 11 block** (`T123–T142`, added 2026-09-02, none of the twenty built, plus `T143`, added 2026-09-03). **Every earlier figure is WITHDRAWN, not restated**, and the sequence is kept because the reason each one died is the lesson:

- *"Task count: 122 … `T001–T122`"* — superseded when Phase 11 added twenty tasks.
- *"94 ticked and 48 not / 28 carried forward"* — superseded 2026-09-03 by T115 and T116.
- *"96 ticked and 46 not / 26 carried forward"* — **it was already wrong when this file was last read**: the tree measured **97/45**. T055 had been ticked without its own note or this block being updated, so the file disagreed with itself in two places at once. That is exactly the failure mode the "re-derive" instruction below exists to catch, and it caught it.
- *"99 ticked and 43 not / 23 carried forward"* — held for roughly an hour. It was written after T079 and T083 were reconciled and superseded by T102 **within the same reconciliation**.
- *"Task count: 142 … 100 ticked and 42 not / 22 carried forward"* — **wrong in two independent ways at once, and neither was a tick.** The TOTAL was already 143 when it was written: `T143` had been added to this file and this block was not updated with it, so the file disagreed with itself about how many tasks it contains — the identical failure the `96/46` bullet above records, one line lower. And the split was then superseded by the T063/T066/T069/T117 ticks below.

The present 104/39 reflects a second reconciliation on 2026-09-03, which ticked **T063** (SC-014 proven over every one of 560 hits, with an `M1-locus` mutation that fails the RUN), **T066** (the new-kinds benchmark, now in `platform/backend/testdata/benchmark/` beside the gate that reads it), **T069** (search filters and locus display, 97/97 unit tests) and **T117** (the limits-completeness gate at rc=0, 15 of 15). **Nothing was unticked to pay for them**, the twenty Phase 11 `[UNBUILT]` tasks were not touched, and `T041` was left entirely alone because another agent was editing it.

Read the `142` bullet as a warning about this whole block, not as trivia — and read T102's as the same warning one turn earlier. T102 was measured `[ ]` with a hardcoded `review=None`, and was fixed, gated and paired-proven **while this file was being edited** — the note written for it was stale before it was saved, and only a re-measurement caught it. The second reconciliation then found **six** notes stale in the same way (T063, T066, T068, T069, T117, T143), every one of them because work landed under them within hours. **A count here is a reading of one moment.** Re-derive rather than trusting the sentence above; this file's checkbox state has now gone stale five times:

```
cd specs/002-knowledge-areas-deep-linking
grep -c '^- \[x\] T' tasks.md    # 104
grep -c '^- \[ \] T' tasks.md    # 39
grep -c '^- \[[ xX]\] T' tasks.md  # 143
grep '^- \[ \] T' tasks.md | grep -c '\[UNBUILT\]'
        # 21 — the 20 Phase 11 OCR tasks plus T143
```